# Supplementary Material for [Experiment, Analysis, and Benchmark] CANTraj: Recovering Vehicle Trajectories from CAN Signals in Naturalistic Driving

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***Abstract*—This document accompanies the paper with the material that supports its reproduction: the structure of the released package and the commands that rebuild every table and figure (S1), the benchmark definitions in full (S2), the reproduction details of the six published methods and of the benchmark baselines (S3), the complete result tables behind the paper's findings, including the figures that the page limit left out (S4), the full protocol, prompts and per-configuration results of the supplementary agent study (S5), and further recovered windows with their camera frames (S6). Code: https://github.com/WangYuHang-cmd/GPS--CAN. Data, scores, configurations, logs and weights: https://huggingface.co/datasets/HenryYHW/CANTraj.**

## I. S1. Reproduction package

The code repository holds the `cantraj` package (data layer, signal whitelist, causal replay, metrics and scoring, baselines, map graphs, learned motion models, the agent study), the scripts that run every experiment and build every table and figure, the frozen configuration (`configs/`: benchmark thresholds and manifest hashes, signal registry with provenance classes, signal groups, protocols, windows, agent prompts and budgets), the specification of benchmark v1 and the test suite (139 tests on synthetic data). The data package holds CANTraj (3,940 drives of 454 drivers: re-decoded CAN event streams at logged times, the aligned GNSS reference with the first and last two minutes of every drive cut, quality metadata and driver-disjoint splits), the benchmark manifests (episodes, map-start rule, tile list with md5 and snapshot date, cohorts), every per-episode score of every run, the run registry with code, scorer and tile hashes, the agent study's configuration and conversation logs, the map tiles and the learned models' weights.

*a) Rebuilding the tables:* From the released per-episode scores alone: copy the package's `tables/` and `reports/` into `reports/icde/`, then run `icde_tables.py --extras a3 pairs topk signal cohort cost robust robust_map`, `icde_area_check.py`, `icde_extra.py` (and `--what supp`), `icde_spy_dewri_diag.py`, `failure_analysis.py`, `icde_figures.py` and `icde_macros.py`. Every number of the paper is a macro in `macros_numbers.tex` written by the last script.

*b) Rerunning the methods:* `run_baseline.py` with the v1.1 flags (`--version v1.1 --manifest episode_manifest_v1.parquet --eligible-only --tiles-dir maps/tiles_v2 --map-episodes episode_map_v11.parquet`), one chain script per study (`scripts/run_icde_*.sh`), then `score.py --run <id>`. Prediction files are regenerated, not shipped. The agent study needs a served Qwen3-14B (revision pinned in the configuration) behind an OpenAI-compatible endpoint; `scripts/h100_setup.sh` installs and serves it, `run_icde_agent.sh` runs the cache, the selector, the three seeds, the scoring and the tables.

## II. S2. Benchmark definitions

*a) Episodes and eligibility:* Windows of 60, 300 and 900 s and whole drives tile each drive from its first reference fix; a window starts at an anchor fix (speed $\geq 3$ m/s, bearing accuracy within 5°, horizontal accuracy under 10 m, CAN speed $\geq 2.5$ m/s) and is eligible when it has at least 0.8 reference fixes per second, no reference gap over 30 s and a path of at least 100 m. Drives with an unconfirmed vehicle platform and the controlled supplement stay out of the main board. The map-start rule of v1.1 requires a drivable edge within 400 m of the start in the visible map; windows that fail it are reported separately and never counted against a method.

*b) Splits:* Driver-disjoint folds: train, validation, test and few-shot held-out (776 held-out drives of 94 drivers). Parameters and models use train and validation only.

*c) Signals and provenance:* Signal sets S1 (speed), S2 (speed and pedals), S3 (speed and steering) and S4 (all); groups G1 to G8 whitelist columns and static parameters; every signal carries a provenance class (bus-read, derived in the vehicle software, logger sensor, GPS-informed), and class-F references are never in the CAN-only board.

*d) Protocols:* Online or offline access, map or no map, start heading given or withheld; the visible map is the disc of radius $5\,\mathrm{km} + 40\,\mathrm{m/s} \cdot D$ (at most 60 km) around the start, enforced per window.

*e) Metrics:* ADE and FDE at the reference seconds, coverage within 10 to 500 m, time-free path recall and precision at 25, 50 and 100 m, route recovery (path recall at 50 m $\geq 0.8$),

terminal excursion and tail-lost flags; an expected window without output is a failure (ADE $+\infty$, rates 0); medians carry driver-clustered bootstrap intervals (1,000 resamples); paired differences are computed on common windows.

## III. S3. Methods

The six published methods were re-implemented from their papers and, where available, checked against the authors' released code on the authors' own samples (OPD-II: 8 of 59 logs within a median of at most 0.33 m of the unmodified script; MacroTracking: the sample trace within a median of 7.1 m of the committed trajectory). Adaptations are named as such in Table I: Elastic Pathing and Dewri et al. with the start heading (start edges within 45° of it), CAN-Trace and SPy with a 5 km square centred on the start, OPD-II with a strictly online read-out. Parameters left open by the papers were set on training and validation drives only. The benchmark baselines are dead reckoning (nominal, calibrated, with an online offset), a route prior and route retrieval, a map particle filter (2,000 particles, 10 Hz ticks, heading and turn noise of 15 and 20°) and the offline map beam search (beam 200, ten candidates, heading and turn widths 15 and 20°, U-turn weight 0.05, re-anchoring gain 0.3, length slack 2 % with a 10 m floor, dead-end weight 0.1, 50 m start radius, 10 m merge bins, temperature 1), as given in Algorithm 1 of the paper. The learned motion models (a GRU and a physics residual, three seeds) are trained on the training fold only with validation early stopping.

## IV. S4. Complete results

## V. S5. Agent study

The supplementary agent study (Section VI of the paper) compares, on one five-minute window per held-out drive (268 test windows of 42 drivers, 391 few-shot windows of 50 drivers), the offline beam (A), a supervised selector over the three reconstructors (B) and a Qwen3-14B tool-calling controller (C). The protocol was frozen after a validation pilot (`reports/agent_protocol_v1.md`): model revision, sampling, prompt, tool schema, budget, selector weights and cohort hashes are recorded there; every window of every seed received a valid choice. Tables XXII to XXV give every row.

*a) Cost:* The reconstructed serial cost of a window is the model's wall time plus the measured single-worker time of the tools it ran plus the tool execution time. On 10 validation windows every reconstructor call was executed uncached and single-stream: median end-to-end time 27.7 s against a reconstructed serial cost of 23.4 s on the same windows (ratio 1.08, 90th percentile 1.68).

*b) System prompt (verbatim):*

```
You are the controller of a vehicle-trajectory recovery
system. One five-minute driving window is given. The
system knows only the vehicle's CAN signals (speed and
steering-wheel angle), the start position (the origin),
the start heading, static vehicle parameters and the road
graph within the visible disc around the start. No GPS
positions, no true route and no scores are available to
you or to the tools; your job is to pick the candidate
path most likely to be the vehicle's true path.

Tools: run_reconstructor(name) runs one of three existing
reconstructors (beam, pf, ep) and returns its ranked
candidate paths with diagnostics;
inspect_candidate(candidate_id) gives the full diagnostics
of a candidate you have already seen;
query_local_graph(candidate_id, distance_m) describes the
road junction nearest to that distance along a candidate
you have seen; submit_choice(candidate_id, reason) ends
the task. Every reconstructor can be run at most once. You
must finish with submit_choice for a candidate id that a
reconstructor returned to you; nothing else is accepted.

Diagnostics (all computed from the CAN signals, the given
start heading and the road graph): path_len_m is the
candidate's length; len_ratio_dev is |path length /
CAN-integrated distance - 1|; hd_abs_rms_deg is the RMS
difference between the candidate's road bearing and the
dead-reckoned heading (start heading plus integrated
steering) over the moving seconds; hd_drift_rms_deg is the
same after removing the best constant offset; turns_path
and turns_dr count turns (bearing change above 45 degrees
within 40 m) along the candidate and in the dead-reckoned
heading, turns_unmatched counts turns present in one but
not the other (same sign, within 50 m of path distance);
uturns counts reversals; overlap_with_called_top1 is the
share of the candidate within 15 m of the Top-1 of another
reconstructor you have run (unknown for reconstructors you
have not run); end_dist_to_called_top1_m compares end
points; n_called_tools_agree counts other run
reconstructors whose Top-1 overlaps this candidate by at
least 0.8. Positions are metres east and north of the
start; compass headings are degrees clockwise from north.
Dead reckoning accumulates error, so small heading or turn
disagreement does not by itself rule out a candidate;
large unmatched turns or a path much shorter or longer
than the CAN distance are strong evidence against it.

Budget: at most 4 decision turns, at most 7 tool calls
before the final submit_choice, and a limited number of
generated tokens. You may request several tool calls in
one turn (for example two reconstructors at once); each
call counts toward the 7. Be concise. When the budget
forces it, you will be asked to submit immediately.
```

*c) Tool schema and budget:*

```
run_reconstructor(name)
Run one trajectory reconstructor on this window (each
reconstructor can be run once; a repeat returns the same
result and still counts). Returns its ranked candidate
paths with observable diagnostics.
inspect_candidate(candidate_id)
Detailed diagnostics of one candidate you have already
seen: heading consistency, turn matching against the
dead-reckoned heading, end position, overlap with the
Top-1 of the other reconstructors you have run, and where
it diverges from them.
query_local_graph(candidate_id, distance_m)
Road junction nearest to a given distance along a
candidate you have seen: arrival heading, the exits
(relative turn, length, road class, one-way), which exit
the candidate took, and the dead-reckoned heading change
around that moment.
submit_choice(candidate_id, reason)
Final answer: the id of one candidate you have seen. The
program outputs that candidate's full trajectory.
budget: 4 decision turns, 7 tool calls + 1 submit, 4096
completion tokens (per turn 2048, reserve 256), virtual
clock 600 s
```

*d) Extended configuration ($B^+$ and $C^+$ in Table IV of the paper):* After the frozen study, the same controller was given ten reconstructor presets (wider and looser beams, four more filter seeds, Elastic Pathing without heading), two further tools (side-by-side comparison with the divergence junction; the selector's probability), dead-reckoning shape and junction-consistency evidence, calibration hints from the validation

TABLE I

METHODS AND THE INFORMATION THEY USE. $p_0$: START POSITION, $\theta_0$: START HEADING. ONLINE METHODS ANSWER EACH QUERY FROM THE SIGNALS UP TO THAT TIME. MAP: "OSM DISC" IS THE EPISODE'S VISIBLE DISC (SECTION II). DR: DEAD RECKONING, PF: PARTICLE FILTER, GRU: GATED RECURRENT UNIT, S1 TO S4: SIGNAL SETS (SECTION III).

| method | source | signals | priors | map | mode | training data | output |
|---|---|---|---|---|---|---|---|
| Hold start | baseline | none | $p_0$ | none | online | none | position |
| Straight line | baseline | speed | $p_0, \theta_0$ | none | online | none | position |
| Steering DR | baseline | speed, steering | $p_0, \theta_0$ | none | online | none / calib.[a] | position |
| Start prior | baseline | speed | $p_0, \theta_0$ | none | online | train GPS (statistics) | position |
| Route retrieval | baseline | speed | $p_0, \theta_0$ | none | online | train GPS (routes) | position |
| Learned GRU | baseline | speed, steering | $p_0, \theta_0$ | none / OSM disc | online | train GPS | position |
| Learned physics residual | baseline | speed, steering | $p_0, \theta_0$ | none / OSM disc | online | train GPS | position |
| Map PF | baseline | speed (+steering) | $p_0$ (+$\theta_0$) | OSM disc | online | none | position |
| Map beam | baseline | speed (+steering) | $p_0$ (+$\theta_0$) | OSM disc | offline | none | top-10 paths |
| MacroTracking [1] | re-impl. | speed, steering | $p_0, \theta_0$ | OSM disc | online | none / calib.[a] | position |
| OPD-II [2] | re-impl. | speed, steering | $p_0, \theta_0$ | OSM, Meili[e] | delayed $\leq$3 s[b] | none / calib.[a] | position |
| Elastic Pathing [3] | re-impl. | speed (1 Hz) | $p_0$[c] (+$\theta_0$) | OSM disc | offline | val (2 values) | top-$n$ paths |
| CAN-Trace [4] | re-impl. | speed, accel. pedal | area[c] / $p_0$ | OSM disc | offline | none | top-10 paths |
| Dewri et al. [5] | re-impl. | speed (1 Hz) | $p_0$[c] (+$\theta_0$) | OSM disc | offline | none | top-10 paths |
| SPy [6] | re-impl. | steering | area[c] / $p_0$ (+$\theta_0$) | OSM disc | offline | val (2 values) | path |
| Signal-set GRU (S1 to S4) | study | per set | $p_0, \theta_0$ | none / OSM disc | online | train GPS | position |
| Curvature DR (class F) | reference | speed, localizer curv. | $p_0, \theta_0$ | none / OSM disc | online | GPS-informed[d] | position |

[a] variant with training-fold steering calibration. [b] each position is fixed when its 3 s batch is map-matched; a strictly online read-out is also reported. [c] native prior first, then our adaptations to the shared protocols; "area" is a 5 km square centred on the known start. [d] the localizer used GPS in the software versions of this data; reference only. [e] Meili matches within its Valhalla region tiles, 200 m around the dead-reckoned trace, and is not cut to the episode's disc.

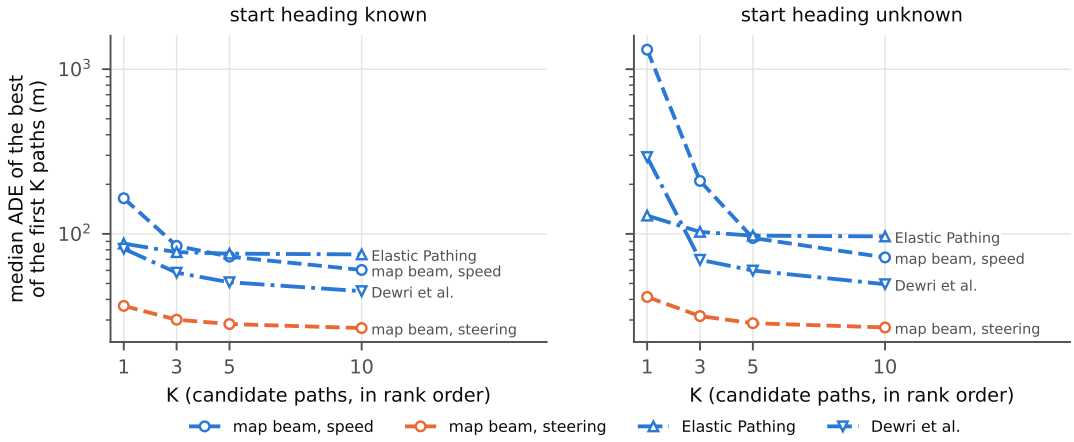



Fig. 1. Best of the first $K$ candidates (whole-path oracle median ADE and per-second union coverage within 100 m) for the offline beam with and without steering and for Elastic Pathing and Dewri et al., five-minute windows.

drives and a relaxed budget (12 turns, 40 calls, 24,576 tokens; the served context was 16,384 tokens); a critic pass and a three-seed vote were added as variants. The configuration was specified after the frozen study from validation data only and run once on the same 659 windows; the rows below also pair it with the frozen configuration. These runs are exploratory and reported in full.

## VI. S6. RECOVERED WINDOWS

## REFERENCES


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[2] T. Bianchi, A. Brighente, M. Conti, and A. Valori, "Your car tells me where you drove: A novel path inference attack via CAN bus and OBD-II data," in *Proceedings of the 10th IEEE European Symposium on Security and Privacy (EuroS&P)*, 2025, pp. 113–132.

[3] X. Gao, B. Firner, S. Sugrim, V. Kaiser-Pendergrast, Y. Yang, and J. Lindqvist, "Transforming speed sequences into road rays on the map with elastic pathing," arXiv:1710.06932, 2017.

[4] X. Lin, B. Ma, X. Wang, G. Yu, Y. He, W. Ni, and R. P. Liu, "CAN-Trace attack: Exploit CAN messages to uncover driving trajectories," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 3, pp. 3223–3236, 2025.

[5] R. Dewri, P. Annadata, W. Eltarjaman, and R. Thurimella, "Inferring trip destinations from driving habits data," in *Proceedings of the 12th ACM Workshop on Privacy in the Electronic Society (WPES)*, 2013, pp. 267–272.

[6] M. D. Pesé, X. Pu, and K. G. Shin, "SPy: Car steering reveals your trip route!" *Proceedings on Privacy Enhancing Technologies*, vol. 2020, no. 2, pp. 155–174, 2020.

TABLE II

COMPLETE BOARD, 5 MIN WINDOWS, BENCHMARK V1.1: EXPECTED WINDOWS $n$, FAILURE SHARE (NO OUTPUT), FAILURE-AWARE MEDIAN ADE IN METRES WITH THE 95 % DRIVER-CLUSTERED CI, AND THE ROUTE RECOVERY RATE (PATH RECALL AT 50 M $\geq$ 0.8), TEST AND FEW-SHOT DRIVERS.

| method | protocol | $n$ | fail % | median ADE m [CI] | route % | $n$ | fail % | median ADE m [CI] | route % |
|---|---|---|---|---|---|---|---|---|---|
| hold start | O-N-Hp | 1,917 | 0.0 | 3,314 [2,113, 4,313] | 0 | 1,963 | 0.0 | 2,709 [2,232, 3,562] | 0 |
| straight line (speed) | O-N-Hp | 1,917 | 0.0 | 735 [644, 911] | 9 | 1,963 | 0.0 | 825 [781, 951] | 7 |
| start prior (speed, train stats) | O-N-Hp | 1,917 | 0.0 | 1,068 [920, 1,119] | 2 | 1,963 | 0.0 | 1,062 [946, 1,162] | 2 |
| route retrieval (speed, train routes) | O-N-Hp | 1,917 | 0.0 | 3,083 [2,002, 4,296] | 4 | 1,963 | 0.0 | 2,530 [2,016, 3,258] | 5 |
| steering DR, nominal | O-N-Hp | 1,917 | 0.0 | 921 [461, 1,742] | 4 | 1,963 | 0.0 | 1,669 [906, 2,472] | 2 |
| steering DR, online offset | O-N-Hp | 1,917 | 0.0 | 869 [611, 1,306] | 3 | 1,963 | 0.0 | 1,423 [964, 1,979] | 2 |
| steering DR, train calibrated | O-N-Hp | 1,917 | 0.0 | 1,312 [409, 4,189] | 4 | 1,963 | 0.0 | 1,651 [919, 2,430] | 2 |
| learned GRU (3 seeds) | O-N-Hp | 1,917 | 0.0 | 543 [226, 2,478] | 7 | 1,963 | 0.0 | 393 [305, 492] | 5 |
| learned physics residual (3 seeds) | O-N-Hp | 1,917 | 0.0 | 690 [310, 2,994] | 6 | 1,963 | 0.0 | 435 [366, 536] | 4 |
| map PF, speed only | O-M-Hp | 1,917 | 0.0 | 286 [50, 868] | 56 | 1,959 | 0.0 | 618 [186, 890] | 48 |
| map PF, steering | O-M-Hp | 1,917 | 0.0 | 43 [29, 59] | 87 | 1,959 | 0.0 | 27 [19, 47] | 90 |
| map PF, learned GRU motion (3 seeds) | O-M-Hp | 1,917 | 0.0 | 89 [42, 168] | 86 | 1,959 | 0.0 | 42 [29, 79] | 86 |
| map PF, learned residual motion (3 seeds) | O-M-Hp | 1,917 | 0.0 | 87 [45, 262] | 86 | 1,959 | 0.0 | 53 [33, 84] | 86 |
| MacroTracking | O-M-Hp | 1,917 | 0.0 | 84 [66, 210] | 62 | 1,959 | 0.0 | 52 [40, 143] | 65 |
| MacroTracking, train calibrated | O-M-Hp | 1,917 | 0.0 | 76 [61, 172] | 65 | 1,959 | 0.0 | 48 [38, 126] | 67 |
| OPD-II, online read-out | O-M-Hp | 1,917 | 0.0 | 42 [27, 69] | 79 | 1,959 | 0.0 | 24 [15, 54] | 78 |
| map beam, speed only | F-M-Hp | 1,917 | 0.0 | 165 [37, 633] | 54 | 1,959 | 0.0 | 343 [88, 637] | 47 |
| map beam, steering | F-M-Hp | 1,917 | 0.0 | 37 [24, 54] | 88 | 1,959 | 0.0 | 21 [17, 40] | 92 |
| map PF, steering, best-lineage read-out | F-M-Hp | 1,917 | 0.0 | 39 [28, 56] | 87 | 1,959 | 0.0 | 24 [17, 44] | 90 |
| Elastic Pathing (start heading used) | F-M-Hp | 1,917 | 4.6 | 87 [38, 324] | 58 | 1,959 | 4.0 | 80 [54, 165] | 59 |
| OPD-II (3 s delayed) | F-M-Hp | 1,917 | 0.0 | 42 [27, 68] | 79 | 1,959 | 0.0 | 24 [15, 54] | 78 |
| Dewri et al. (start heading used) | F-M-Hp | 1,917 | 0.9 | 81 [38, 215] | 62 | 1,959 | 2.8 | 89 [59, 157] | 59 |
| SPy, start known (start heading used) | F-M-Hp | 1,917 | 0.3 | 1,474 [1,283, 1,602] | 20 | 1,959 | 0.2 | 1,284 [1,090, 1,596] | 22 |
| map PF, steering | O-M-Hm | 1,917 | 0.0 | 138 [99, 175] | 79 | 1,959 | 0.0 | 88 [61, 124] | 82 |
| map PF, speed only | O-M-Hm | 1,917 | 0.0 | 1,773 [1,550, 1,987] | 34 | 1,959 | 0.0 | 1,868 [1,572, 2,097] | 29 |
| map beam, steering | F-M-Hm | 1,917 | 0.0 | 41 [26, 59] | 81 | 1,959 | 0.0 | 23 [17, 42] | 87 |
| map beam, speed only | F-M-Hm | 1,917 | 0.0 | 1,314 [866, 1,619] | 32 | 1,959 | 0.0 | 1,527 [1,294, 1,738] | 29 |
| Elastic Pathing | F-M-Hm | 1,917 | 3.2 | 129 [42, 691] | 53 | 1,959 | 2.1 | 177 [94, 344] | 50 |
| CAN-Trace (pedal subset), start known | F-M-Hm | 245 | 99.2 | fail [–, –] | 0 | 162 | 96.9 | fail [–, –] | 0 |
| CAN-Trace (pedal subset), 5 km area | F-M-Hm | 245 | 65.3 | fail [4,344, –] | 0 | 162 | 60.5 | fail [–, –] | 0 |
| Dewri et al. | F-M-Hm | 1,917 | 0.4 | 293 [81, 775] | 46 | 1,959 | 1.3 | 487 [272, 708] | 43 |
| SPy, start known | F-M-Hm | 1,917 | 0.1 | 2,198 [1,702, 2,636] | 11 | 1,959 | 0.1 | 1,810 [1,512, 2,211] | 13 |
| SPy, 5 km area | F-M-Hm | 1,917 | 11.8 | 3,929 [3,222, 4,406] | 0 | 1,959 | 5.5 | 3,400 [3,050, 3,953] | 0 |
| curvature DR (class F, not CAN-only) | O-N-Hp | 1,917 | 0.0 | 84 [74, 107] | 27 | 1,963 | 0.0 | 80 [70, 92] | 27 |
| map PF, curvature (class F) | O-M-Hp | 1,917 | 0.0 | 44 [29, 61] | 86 | 1,959 | 0.0 | 28 [19, 49] | 89 |

Left block: test drivers; right block: few-shot drivers. Protocols O/F (online / offline), N/M (no map / map), Hp/Hm (start heading given / withheld).

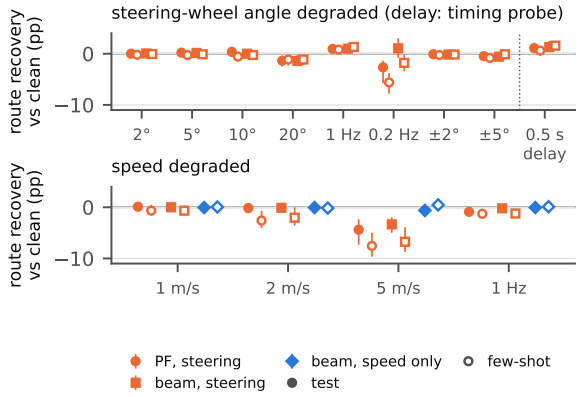



Fig. 2. Fixed map attacks on degraded inputs: change in the five-minute route recovery rate against the clean signal on the same windows (percentage points; 95 % driver-clustered CIs; filled: test, hollow: few-shot drivers); every level of the grid fixed before the first run; the 0.5 s delay is a timing probe.

TABLE III

COMPLETE BOARD, 15 MIN WINDOWS, BENCHMARK V1.1: EXPECTED WINDOWS $n$, FAILURE SHARE (NO OUTPUT), FAILURE-AWARE MEDIAN ADE IN METRES WITH THE 95 % DRIVER-CLUSTERED CI, AND THE ROUTE RECOVERY RATE (PATH RECALL AT 50 M $\geq 0.8$), TEST AND FEW-SHOT DRIVERS.

| method | protocol | $n$ | fail % | median ADE m [CI] | route % | $n$ | fail % | median ADE m [CI] | route % |
|---|---|---|---|---|---|---|---|---|---|
| hold start | O-N-Hp | 608 | 0.0 | 8,759 [5,761, 12,194] | 0 | 565 | 0.0 | 7,494 [5,973, 10,196] | 0 |
| straight line (speed) | O-N-Hp | 608 | 0.0 | 3,389 [3,124, 4,037] | 1 | 565 | 0.0 | 3,906 [3,630, 4,335] | 1 |
| start prior (speed, train stats) | O-N-Hp | 608 | 0.0 | 3,601 [3,128, 3,841] | 0 | 565 | 0.0 | 3,775 [3,545, 4,116] | 0 |
| route retrieval (speed, train routes) | O-N-Hp | 608 | 0.0 | 8,365 [5,617, 12,123] | 1 | 565 | 0.0 | 7,326 [5,648, 9,704] | 3 |
| steering DR, nominal | O-N-Hp | 608 | 0.0 | 5,135 [2,276, 7,673] | 1 | 565 | 0.0 | 7,174 [5,078, 9,162] | 0 |
| steering DR, online offset | O-N-Hp | 608 | 0.0 | 4,805 [3,893, 6,281] | 1 | 565 | 0.0 | 6,570 [5,331, 8,340] | 0 |
| steering DR, train calibrated | O-N-Hp | 608 | 0.0 | 6,375 [2,069, 12,288] | 1 | 565 | 0.0 | 7,138 [5,117, 9,181] | 0 |
| learned GRU (3 seeds) | O-N-Hp | 608 | 0.0 | 3,010 [1,193, 10,461] | 1 | 565 | 0.0 | 2,569 [1,923, 3,128] | 0 |
| learned physics residual (3 seeds) | O-N-Hp | 608 | 0.0 | 4,799 [1,785, 10,746] | 2 | 565 | 0.0 | 2,834 [2,292, 3,635] | 0 |
| map PF, speed only | O-M-Hp | 608 | 0.0 | 2,326 [364, 4,401] | 37 | 561 | 0.0 | 3,444 [2,231, 4,825] | 27 |
| map PF, steering | O-M-Hp | 608 | 0.0 | 178 [85, 373] | 69 | 561 | 0.0 | 127 [52, 224] | 73 |
| MacroTracking | O-M-Hp | 608 | 0.0 | 851 [293, 3,183] | 40 | 561 | 0.0 | 1,064 [245, 4,190] | 39 |
| MacroTracking, train calibrated | O-M-Hp | 608 | 0.0 | 779 [284, 2,875] | 39 | 561 | 0.0 | 1,110 [155, 3,912] | 40 |
| OPD-II, online read-out | O-M-Hp | 608 | 0.0 | 223 [100, 963] | 55 | 561 | 0.0 | 497 [94, 2,073] | 47 |
| map beam, speed only | F-M-Hp | 608 | 0.0 | 2,142 [136, 4,651] | 40 | 561 | 0.0 | 3,084 [1,498, 4,609] | 30 |
| map beam, steering | F-M-Hp | 608 | 0.0 | 149 [68, 331] | 73 | 561 | 0.0 | 106 [45, 174] | 76 |
| map PF, steering, best-lineage read-out | F-M-Hp | 608 | 0.0 | 165 [79, 394] | 68 | 561 | 0.0 | 125 [46, 215] | 72 |
| Elastic Pathing (start heading used) | F-M-Hp | 608 | 9.2 | 1,094 [124, 3,263] | 42 | 561 | 8.4 | 1,223 [928, 1,861] | 42 |
| OPD-II (3 s delayed) | F-M-Hp | 608 | 0.0 | 223 [100, 962] | 55 | 561 | 0.0 | 497 [94, 2,072] | 47 |
| Dewri et al. (start heading used) | F-M-Hp | 608 | 5.3 | 1,053 [142, 2,985] | 44 | 561 | 9.1 | 2,058 [1,159, 2,764] | 37 |
| SPy, start known (start heading used) | F-M-Hp | 608 | 0.5 | 5,396 [4,436, 6,077] | 6 | 561 | 0.5 | 4,819 [4,176, 6,440] | 6 |
| map PF, steering | O-M-Hm | 608 | 0.0 | 340 [201, 706] | 63 | 561 | 0.0 | 242 [118, 481] | 68 |
| map PF, speed only | O-M-Hm | 608 | 0.0 | 5,799 [4,754, 6,909] | 15 | 561 | 0.0 | 5,901 [4,652, 6,912] | 10 |
| map beam, steering | F-M-Hm | 608 | 0.0 | 155 [68, 376] | 70 | 561 | 0.0 | 110 [46, 184] | 74 |
| map beam, speed only | F-M-Hm | 608 | 0.0 | 5,099 [3,859, 6,250] | 25 | 561 | 0.0 | 5,419 [3,588, 6,052] | 21 |
| Elastic Pathing | F-M-Hm | 608 | 6.6 | 1,214 [135, 3,777] | 42 | 561 | 5.3 | 1,602 [1,001, 2,136] | 40 |
| CAN-Trace (pedal subset), 5 km area | F-M-Hm | 79 | 93.7 | fail [–, –] | 0 | 45 | 95.6 | fail [–, –] | 0 |
| Dewri et al. | F-M-Hm | 608 | 2.5 | 3,064 [1,642, 4,239] | 32 | 561 | 4.8 | 3,700 [2,838, 4,731] | 27 |
| SPy, start known | F-M-Hm | 608 | 0.2 | 6,199 [4,874, 7,741] | 4 | 561 | 0.4 | 5,928 [4,883, 7,689] | 4 |
| SPy, 5 km area | F-M-Hm | 608 | 6.6 | 9,871 [6,632, 12,216] | 0 | 561 | 4.8 | 8,349 [6,967, 10,942] | 0 |
| curvature DR (class F, not CAN-only) | O-N-Hp | 608 | 0.0 | 511 [445, 644] | 4 | 565 | 0.0 | 491 [424, 595] | 2 |
| map PF, curvature (class F) | O-M-Hp | 608 | 0.0 | 186 [86, 430] | 67 | 561 | 0.0 | 138 [61, 226] | 71 |

Left block: test drivers; right block: few-shot drivers. Protocols O/F (online / offline), N/M (no map / map), Hp/Hm (start heading given / withheld).

TABLE IV

COMPLETE BOARD, WHOLE DRIVE WINDOWS, BENCHMARK V1.1: EXPECTED WINDOWS $n$, FAILURE SHARE (NO OUTPUT), FAILURE-AWARE MEDIAN ADE IN METRES WITH THE 95 % DRIVER-CLUSTERED CI, AND THE ROUTE RECOVERY RATE (PATH RECALL AT 50 M $\geq 0.8$), TEST AND FEW-SHOT DRIVERS.

| method | protocol | $n$ | fail % | median ADE m [CI] | route % | $n$ | fail % | median ADE m [CI] | route % |
|---|---|---|---|---|---|---|---|---|---|
| hold start | O-N-Hp | 282 | 0.0 | 5,235 [3,825, 7,854] | 0 | 460 | 0.0 | 3,322 [2,580, 4,870] | 0 |
| straight line (speed) | O-N-Hp | 282 | 0.0 | 6,551 [4,548, 9,073] | 2 | 460 | 0.0 | 3,327 [2,389, 4,744] | 8 |
| start prior (speed, train stats) | O-N-Hp | 282 | 0.0 | 5,771 [3,750, 7,481] | 2 | 460 | 0.0 | 2,912 [2,054, 3,825] | 6 |
| route retrieval (speed, train routes) | O-N-Hp | 282 | 0.0 | 5,586 [3,843, 7,854] | 1 | 460 | 0.0 | 3,236 [2,360, 4,800] | 3 |
| steering DR, nominal | O-N-Hp | 282 | 0.0 | 2,708 [1,468, 8,341] | 5 | 460 | 0.0 | 2,674 [1,251, 3,916] | 7 |
| steering DR, online offset | O-N-Hp | 282 | 0.0 | 5,004 [3,266, 6,809] | 4 | 460 | 0.0 | 3,118 [1,868, 4,449] | 8 |
| steering DR, train calibrated | O-N-Hp | 282 | 0.0 | 3,121 [1,917, 8,403] | 4 | 460 | 0.0 | 2,656 [1,498, 3,902] | 7 |
| learned GRU (3 seeds) | O-N-Hp | 282 | 0.0 | 1,704 [1,149, 4,563] | 6 | 460 | 0.0 | 887 [562, 1,547] | 11 |
| learned physics residual (3 seeds) | O-N-Hp | 282 | 0.0 | 1,944 [1,255, 5,372] | 6 | 460 | 0.0 | 1,100 [559, 1,838] | 10 |
| map PF, speed only | O-M-Hp | 282 | 0.0 | 5,827 [3,211, 7,147] | 8 | 454 | 0.0 | 2,908 [1,996, 3,946] | 13 |
| map PF, steering | O-M-Hp | 282 | 0.0 | 1,906 [799, 3,128] | 33 | 454 | 0.0 | 193 [91, 741] | 57 |
| MacroTracking | O-M-Hp | 282 | 0.0 | 1,649 [779, 4,479] | 33 | 454 | 0.0 | 227 [73, 900] | 46 |
| OPD-II, online read-out | O-M-Hp | 282 | 0.0 | 1,712 [913, 4,120] | 35 | 454 | 0.0 | 407 [52, 1,231] | 50 |
| map beam, speed only | F-M-Hp | 282 | 0.0 | 5,930 [4,134, 8,365] | 8 | 454 | 0.0 | 3,095 [2,044, 3,941] | 12 |
| map beam, steering | F-M-Hp | 282 | 0.0 | 1,179 [343, 1,729] | 41 | 454 | 0.0 | 94 [44, 239] | 61 |
| Elastic Pathing (start heading used) | F-M-Hp | 282 | 11.0 | 3,597 [1,952, 5,600] | 17 | 454 | 10.1 | 1,430 [957, 2,343] | 29 |
| OPD-II (3 s delayed) | F-M-Hp | 282 | 0.0 | 1,711 [912, 4,120] | 35 | 454 | 0.0 | 407 [53, 1,231] | 50 |
| Dewri et al. (start heading used) | F-M-Hp | 282 | 13.1 | 5,382 [2,969, 7,392] | 13 | 454 | 15.0 | 1,758 [1,119, 3,596] | 22 |
| SPy, start known (start heading used) | F-M-Hp | 282 | 1.4 | 4,752 [3,473, 7,599] | 4 | 454 | 0.9 | 2,651 [1,830, 4,240] | 6 |
| map PF, steering | O-M-Hm | 282 | 0.0 | 2,239 [1,076, 3,319] | 31 | 454 | 0.0 | 432 [230, 938] | 51 |
| map PF, speed only | O-M-Hm | 282 | 0.0 | 6,313 [3,834, 8,070] | 4 | 454 | 0.0 | 3,530 [2,312, 4,912] | 9 |
| map beam, steering | F-M-Hm | 282 | 0.0 | 1,244 [447, 2,085] | 39 | 454 | 0.0 | 124 [48, 510] | 58 |
| map beam, speed only | F-M-Hm | 282 | 0.0 | 6,958 [3,931, 9,102] | 6 | 454 | 0.0 | 3,605 [2,503, 4,779] | 9 |
| Elastic Pathing | F-M-Hm | 282 | 6.7 | 3,554 [2,359, 5,631] | 15 | 454 | 7.9 | 1,644 [1,056, 2,585] | 25 |
| Dewri et al. | F-M-Hm | 282 | 8.9 | 5,619 [3,273, 7,780] | 11 | 454 | 10.6 | 2,172 [1,429, 3,600] | 19 |
| SPy, start known | F-M-Hm | 282 | 1.8 | 4,932 [3,613, 7,779] | 2 | 454 | 1.1 | 3,254 [2,245, 4,387] | 3 |
| curvature DR (class F, not CAN-only) | O-N-Hp | 282 | 0.0 | 678 [498, 1,427] | 12 | 460 | 0.0 | 325 [204, 519] | 21 |

Left block: test drivers; right block: few-shot drivers. Protocols O/F (online / offline), N/M (no map / map), Hp/Hm (start heading given / withheld).

TABLE V

SAME-COHORT HORIZON STUDY: ONE START PER DRIVE, SCORED 1, 5 AND 15 MINUTES AFTER THE START (MEDIAN ADE IN METRES / ROUTE RECOVERY %). ONLINE METHODS ON PREFIXES OF THEIR 15-MINUTE RUNS; OFFLINE METHODS RERUN ON EACH PREFIX.

| method | protocol | 1 min | 5 min | 15 min | 1 min | 5 min | 15 min |
|---|---|---|---|---|---|---|---|
| dr_curvature | O-N-Hp | 13 / 81 | 101 / 18 | 569 / 4 | 13 / 91 | 105 / 19 | 577 / 3 |
| dr_none | O-N-Hm | 382 / 4 | 2,150 / 0 | 7,693 / 0 | 439 / 7 | 2,749 / 0 | 8,806 / 0 |
| dr_none | O-N-Hp | 103 / 28 | 1,364 / 1 | 5,317 / 0 | 82 / 31 | 1,141 / 5 | 5,131 / 1 |
| dr_steer | O-N-Hm | 399 / 6 | 2,087 / 0 | 6,610 / 0 | 415 / 6 | 2,019 / 0 | 5,949 / 0 |
| dr_steer | O-N-Hp | 24 / 60 | 500 / 4 | 3,308 / 0 | 40 / 48 | 941 / 3 | 5,066 / 0 |
| dr_steer_offonline | O-N-Hm | 371 / 5 | 1,904 / 0 | 5,618 / 0 | 425 / 5 | 2,108 / 0 | 6,367 / 0 |
| dr_steer_offonline | O-N-Hp | 36 / 46 | 1,080 / 2 | 5,066 / 0 | 37 / 45 | 1,174 / 1 | 5,548 / 0 |
| learned_gru_s0 | O-N-Hm | 385 / 7 | 2,189 / 0 | 7,032 / 0 | 426 / 6 | 2,331 / 0 | 6,733 / 0 |
| learned_gru_s0 | O-N-Hp | 20 / 66 | 282 / 10 | 1,483 / 2 | 25 / 61 | 301 / 4 | 1,989 / 0 |
| learned_gru_s1 | O-N-Hm | 388 / 7 | 2,255 / 0 | 6,927 / 0 | 424 / 7 | 2,361 / 1 | 6,740 / 0 |
| learned_gru_s1 | O-N-Hp | 18 / 67 | 234 / 7 | 1,239 / 1 | 19 / 66 | 289 / 6 | 1,896 / 0 |
| learned_gru_s2 | O-N-Hm | 384 / 6 | 2,207 / 0 | 7,107 / 0 | 425 / 7 | 2,260 / 0 | 6,884 / 0 |
| learned_gru_s2 | O-N-Hp | 19 / 68 | 300 / 7 | 1,438 / 2 | 21 / 67 | 308 / 5 | 2,008 / 0 |
| macrotracking | O-M-Hp | 20 / 83 | 62 / 59 | 1,351 / 35 | 20 / 93 | 46 / 62 | 953 / 37 |
| map_pf_curvature | O-M-Hm | 94 / 72 | 182 / 61 | 707 / 47 | 64 / 78 | 95 / 69 | 662 / 51 |
| map_pf_curvature | O-M-Hp | 19 / 81 | 65 / 63 | 708 / 47 | 16 / 93 | 35 / 75 | 232 / 56 |
| map_pf_none | O-M-Hm | 271 / 44 | 1,309 / 18 | 5,180 / 7 | 278 / 44 | 1,506 / 24 | 5,191 / 7 |
| map_pf_none | O-M-Hp | 113 / 56 | 1,145 / 21 | 4,720 / 12 | 92 / 60 | 1,037 / 31 | 4,383 / 12 |
| map_pf_steer | O-M-Hm | 90 / 74 | 205 / 60 | 815 / 47 | 51 / 82 | 75 / 72 | 483 / 54 |
| map_pf_steer | O-M-Hp | 19 / 82 | 56 / 64 | 672 / 49 | 15 / 93 | 31 / 77 | 249 / 56 |
| opd2_online | O-M-Hp | 11 / 88 | 48 / 61 | 1,316 / 39 | 9 / 92 | 27 / 74 | 867 / 39 |
| elastic_pathing | F-M-Hm | 70 / 49 | 703 / 34 | 3,335 / 20 | 29 / 55 | 316 / 41 | 2,379 / 28 |
| elastic_pathing_hp | F-M-Hp | 23 / 59 | 505 / 37 | 3,437 / 20 | 14 / 72 | 238 / 42 | 2,267 / 29 |
| map_beam_none | F-M-Hm | 167 / 31 | 1,274 / 14 | 5,352 / 11 | 212 / 34 | 1,358 / 27 | 4,763 / 11 |
| map_beam_none | F-M-Hp | 97 / 41 | 1,030 / 18 | 5,107 / 14 | 56 / 51 | 941 / 33 | 3,798 / 13 |
| map_beam_steer | F-M-Hm | 21 / 70 | 44 / 71 | 297 / 58 | 14 / 79 | 27 / 79 | 126 / 62 |
| map_beam_steer | F-M-Hp | 17 / 81 | 44 / 70 | 303 / 58 | 12 / 94 | 25 / 84 | 123 / 63 |
| map_pf_steer_lineage | F-M-Hp | 17 / 81 | 53 / 66 | 675 / 47 | 13 / 93 | 27 / 77 | 241 / 55 |
| opd2 | F-M-Hp | 11 / 88 | 48 / 62 | 1,316 / 38 | 9 / 91 | 26 / 73 | 867 / 39 |
| can_trace_area | F-M-Hm | – | – | fail / 0 | – | – | fail / 0 |
| dewri | F-M-Hm | – | – | 3,312 / 20 | – | – | 3,700 / 18 |
| dewri_hp | F-M-Hp | – | – | 3,069 / 25 | – | – | 3,562 / 22 |
| dr_can_yaw | O-N-Hm | – | – | fail / 0 | – | – | fail / 0 |
| dr_can_yaw | O-N-Hp | – | – | fail / 0 | – | – | fail / 0 |
| dr_imu_yaw_rls | O-N-Hm | – | – | fail / 0 | – | – | fail / 0 |
| dr_imu_yaw_rls | O-N-Hp | – | – | fail / 0 | – | – | fail / 0 |
| dr_none_pn05 | O-N-Hm | – | – | 7,695 / 0 | – | – | 8,806 / 0 |
| dr_none_pn05 | O-N-Hp | – | – | 5,318 / 0 | – | – | 5,145 / 1 |
| dr_none_pq1 | O-N-Hm | – | – | 7,694 / 0 | – | – | 8,816 / 0 |
| dr_none_pq1 | O-N-Hp | – | – | 5,306 / 0 | – | – | 5,116 / 1 |
| dr_none_s1hz | O-N-Hm | – | – | 7,693 / 0 | – | – | 8,806 / 0 |
| dr_none_s1hz | O-N-Hp | – | – | 5,317 / 0 | – | – | 5,131 / 1 |
| dr_steer_calib | O-N-Hm | – | – | 6,987 / 0 | – | – | 6,116 / 0 |
| dr_steer_calib | O-N-Hp | – | – | 3,138 / 1 | – | – | 5,052 / 0 |
| dr_steer_pn05 | O-N-Hm | – | – | 6,605 / 0 | – | – | 5,980 / 0 |
| dr_steer_pn05 | O-N-Hp | – | – | 3,286 / 1 | – | – | 5,047 / 0 |
| dr_steer_pq1 | O-N-Hm | – | – | 6,653 / 0 | – | – | 5,969 / 0 |
| dr_steer_pq1 | O-N-Hp | – | – | 3,473 / 0 | – | – | 5,069 / 0 |
| dr_steer_s1hz | O-N-Hm | – | – | 6,755 / 0 | – | – | 6,120 / 0 |
| dr_steer_s1hz | O-N-Hp | – | – | 3,278 / 2 | – | – | 4,939 / 0 |
| learned_resid_nogru_s0 | O-N-Hm | – | – | 7,632 / 0 | – | – | 6,674 / 0 |
| learned_resid_nogru_s0 | O-N-Hp | – | – | 3,898 / 0 | – | – | 4,718 / 0 |
| learned_resid_nogru_s1 | O-N-Hm | – | – | 7,525 / 0 | – | – | 6,882 / 0 |
| learned_resid_nogru_s1 | O-N-Hp | – | – | 3,792 / 0 | – | – | 5,024 / 0 |
| learned_resid_nogru_s2 | O-N-Hm | – | – | 7,774 / 0 | – | – | 6,494 / 0 |
| learned_resid_nogru_s2 | O-N-Hp | – | – | 3,719 / 0 | – | – | 4,664 / 0 |
| learned_resid_s0 | O-N-Hm | – | – | 6,780 / 0 | – | – | 6,760 / 0 |
| learned_resid_s0 | O-N-Hp | – | – | 2,253 / 2 | – | – | 2,285 / 0 |
| learned_resid_s1 | O-N-Hm | – | – | 7,090 / 0 | – | – | 6,557 / 0 |
| learned_resid_s1 | O-N-Hp | – | – | 1,910 / 2 | – | – | 2,050 / 0 |
| learned_resid_s2 | O-N-Hm | – | – | 6,743 / 0 | – | – | 6,903 / 0 |
| learned_resid_s2 | O-N-Hp | – | – | 1,803 / 2 | – | – | 2,582 / 0 |
| macrotracking_calib | O-M-Hp | – | – | 1,076 / 35 | – | – | 985 / 39 |
| macrotracking_wrapfix | O-M-Hp | – | – | 869 / 36 | – | – | 803 / 38 |
| opd2_calib | F-M-Hp | – | – | 1,175 / 41 | – | – | 970 / 41 |
| opd2_wrapfix | F-M-Hp | – | – | 1,316 / 38 | – | – | 867 / 39 |
| p0_hold | O-N-Hm | – | – | 5,432 / 0 | – | – | 5,512 / 0 |
| p0_hold | O-N-Hp | – | – | 5,432 / 0 | – | – | 5,512 / 0 |
| prior | O-N-Hm | – | – | 5,432 / 0 | – | – | 5,512 / 0 |
| prior | O-N-Hp | – | – | 4,391 / 0 | – | – | 4,328 / 0 |
| retrieval | O-N-Hm | – | – | 5,270 / 0 | – | – | 5,509 / 2 |
| retrieval | O-N-Hp | – | – | 5,270 / 0 | – | – | 5,509 / 2 |
| spy | F-M-Hm | – | – | 5,110 / 1 | – | – | 4,900 / 2 |
| spy_area | F-M-Hm | – | – | 5,917 / 0 | – | – | 5,765 / 0 |
| spy_hp | F-M-Hp | – | – | 4,761 / 2 | – | – | 4,592 / 2 |

Left: test drives; right: few-shot drives.

TABLE VI

SIGNAL STUDY (PEDAL-VALID FIVE-MINUTE WINDOWS OF THE HELD-OUT DRIVES): ONE LEARNER, ONE DECODER, PER SIGNAL SET AND SEED; FAILURE-AWARE MEDIAN ADE WITH 95 % DRIVER-CLUSTERED CI, ROUTE RECOVERY AND 100 M COVERAGE.

| setting | set | seed | $n$ | fail | median ADE m [CI] | route % | c100 % |
|---|---|---|---|---|---|---|---|
| no map | S1 | 0 | 407 | 0 | 1,079 [974, 1,209] | 1 | 19 |
| no map | S1 | 1 | 407 | 0 | 1,182 [1,005, 1,413] | 1 | 18 |
| no map | S1 | 2 | 407 | 0 | 1,128 [1,002, 1,324] | 1 | 19 |
| no map | S2c | 0 | 407 | 0 | 1,156 [1,006, 1,383] | 1 | 19 |
| no map | S2c | 1 | 407 | 0 | 1,156 [1,043, 1,323] | 0 | 18 |
| no map | S2c | 2 | 407 | 0 | 1,028 [854, 1,320] | 2 | 21 |
| no map | S3 | 0 | 407 | 0 | 960 [289, 1,437] | 7 | 29 |
| no map | S3 | 1 | 407 | 0 | 375 [226, 639] | 6 | 36 |
| no map | S3 | 2 | 407 | 0 | 376 [238, 728] | 6 | 35 |
| no map | S4c | 0 | 407 | 0 | 922 [312, 1,242] | 5 | 29 |
| no map | S4c | 1 | 407 | 0 | 401 [232, 759] | 8 | 38 |
| no map | S4c | 2 | 407 | 0 | 378 [274, 595] | 7 | 36 |
| no map | S2b | 0 | 407 | 0 | 1,119 [992, 1,369] | 1 | 19 |
| no map | S4b | 0 | 407 | 0 | 768 [264, 1,182] | 5 | 30 |
| map | S1 | 0 | 407 | 0 | 284 [169, 548] | 50 | 46 |
| map | S1 | 1 | 407 | 0 | 421 [332, 517] | 48 | 39 |
| map | S1 | 2 | 407 | 0 | 312 [201, 505] | 51 | 39 |
| map | S2c | 0 | 407 | 0 | 367 [290, 507] | 50 | 42 |
| map | S2c | 1 | 407 | 0 | 382 [294, 535] | 50 | 43 |
| map | S2c | 2 | 407 | 0 | 490 [394, 702] | 42 | 44 |
| map | S3 | 0 | 407 | 0 | 48 [33, 61] | 87 | 82 |
| map | S3 | 1 | 407 | 0 | 39 [29, 67] | 88 | 85 |
| map | S3 | 2 | 407 | 0 | 58 [25, 73] | 88 | 78 |
| map | S4c | 0 | 407 | 0 | 36 [28, 52] | 86 | 86 |
| map | S4c | 1 | 407 | 0 | 42 [36, 62] | 88 | 85 |
| map | S4c | 2 | 407 | 0 | 47 [34, 60] | 89 | 84 |
| map | S2b | 0 | 407 | 0 | 351 [246, 513] | 51 | 38 |
| map | S4b | 0 | 407 | 0 | 37 [28, 57] | 87 | 86 |

TABLE VII

SIGNAL STUDY, PAIRED GAINS OF ADDING A SIGNAL (A − B, METRES; NEGATIVE = BETTER): MEDIAN AND MEAN PAIRED DIFFERENCES WITH 95 % CIS AND THE SHARE OF WINDOWS THAT IMPROVE.

| setting | added | a | b | seed | $n$ | median diff [CI] | mean diff [CI] | improved % |
|---|---|---|---|---|---|---|---|---|
| no map | pedals — speed | S2c | S1 | 0 | 407 | 42.9 [14.7, 70.6] | 248 [107, 406] | 41 |
| no map | pedals — speed | S2c | S1 | 1 | 407 | -2.3 [-42.8, 54.0] | -159 [-257, 27] | 50 |
| no map | pedals — speed | S2c | S1 | 2 | 407 | -4.1 [-36.9, 20.1] | -81 [-233, 28] | 51 |
| no map | steering — speed | S3 | S1 | 0 | 407 | -134.5 [-515.2, 130.2] | 25 [-614, 358] | 56 |
| no map | steering — speed | S3 | S1 | 1 | 407 | -733.9 [-905.2, -352.2] | -853 [-1,110, -359] | 81 |
| no map | steering — speed | S3 | S1 | 2 | 407 | -530.6 [-747.3, -251.7] | -734 [-963, -288] | 75 |
| no map | pedals — speed+steering | S4c | S3 | 0 | 407 | -25.2 [-58.1, 14.7] | -85 [-135, 15] | 58 |
| no map | pedals — speed+steering | S4c | S3 | 1 | 407 | -3.7 [-22.3, 36.8] | -28 [-177, 110] | 52 |
| no map | pedals — speed+steering | S4c | S3 | 2 | 407 | -12.5 [-67.7, 45.8] | -58 [-219, 64] | 54 |
| no map | steering — speed+pedals | S4c | S2c | 0 | 407 | -258.1 [-658.4, -93.8] | -308 [-888, -21] | 63 |
| no map | steering — speed+pedals | S4c | S2c | 1 | 407 | -587.1 [-720.9, -419.4] | -722 [-898, -365] | 83 |
| no map | steering — speed+pedals | S4c | S2c | 2 | 407 | -405.2 [-652.0, -224.1] | -711 [-871, -413] | 75 |
| no map | pressed flags — speed | S2b | S1 | 0 | 407 | 76.2 [32.6, 102.2] | 162 [83, 279] | 40 |
| no map | pressed flags — speed+steering | S4b | S3 | 0 | 407 | -93.1 [-271.2, 5.0] | -218 [-325, -35] | 70 |
| map | pedals — speed | S2c | S1 | 0 | 407 | 0.9 [-1.6, 7.9] | 13 [-22, 40] | 48 |
| map | pedals — speed | S2c | S1 | 1 | 407 | -4.8 [-9.4, 0.6] | -2 [-48, 104] | 56 |
| map | pedals — speed | S2c | S1 | 2 | 407 | 6.1 [-6.6, 19.7] | 150 [-2, 235] | 44 |
| map | steering — speed | S3 | S1 | 0 | 407 | -119.5 [-305.3, -80.9] | -698 [-893, -581] | 83 |
| map | steering — speed | S3 | S1 | 1 | 407 | -233.7 [-363.5, -149.9] | -727 [-856, -582] | 87 |
| map | steering — speed | S3 | S1 | 2 | 407 | -147.0 [-343.4, -103.8] | -690 [-895, -582] | 88 |
| map | pedals — speed+steering | S4c | S3 | 0 | 407 | -6.4 [-10.9, -0.3] | -13 [-29, 20] | 72 |
| map | pedals — speed+steering | S4c | S3 | 1 | 407 | 0.9 [-1.2, 3.7] | 16 [-14, 55] | 44 |
| map | pedals — speed+steering | S4c | S3 | 2 | 407 | -3.7 [-9.0, 3.8] | -14 [-40, 8] | 63 |
| map | steering — speed+pedals | S4c | S2c | 0 | 407 | -191.3 [-320.7, -144.2] | -724 [-890, -583] | 90 |
| map | steering — speed+pedals | S4c | S2c | 1 | 407 | -231.4 [-348.8, -127.1] | -709 [-894, -570] | 84 |
| map | steering — speed+pedals | S4c | S2c | 2 | 407 | -359.2 [-461.9, -207.2] | -855 [-1,030, -621] | 85 |
| map | pressed flags — speed | S2b | S1 | 0 | 407 | 5.2 [0.8, 10.7] | 14 [-20, 77] | 41 |
| map | pressed flags — speed+steering | S4b | S3 | 0 | 407 | -3.6 [-12.2, 0.4] | -30 [-59, 2] | 67 |

TABLE VIII

CANDIDATE SETS ON FIVE-MINUTE WINDOWS: WHOLE-PATH ORACLE MEDIAN ADE AMONG THE FIRST $K$ CANDIDATES, SHARE OF WINDOWS WHOSE BEST CANDIDATE IS RANKED FIRST, AND THE MEDIAN NUMBER OF CANDIDATES.

| method | protocol | fold | $n$ | out | single | $K$=1 | 3 | 5 | 10 | best 1st % | cands |
|---|---|---|---|---|---|---|---|---|---|---|---|
| map_beam_steer | F-M-Hp | test | 1,917 | 1,917 | 31 | 37 | 30 | 28 | 27 | 58 | 10 |
| map_beam_steer | F-M-Hp | few-shot | 1,959 | 1,959 | 15 | 21 | 18 | 18 | 17 | 61 | 10 |
| map_beam_steer | F-M-Hm | test | 1,917 | 1,917 | 2 | 41 | 32 | 29 | 27 | 54 | 10 |
| map_beam_steer | F-M-Hm | few-shot | 1,959 | 1,959 | 0 | 23 | 19 | 18 | 17 | 58 | 10 |
| map_beam_none | F-M-Hp | test | 1,917 | 1,917 | 31 | 165 | 85 | 73 | 60 | 35 | 10 |
| map_beam_none | F-M-Hp | few-shot | 1,959 | 1,959 | 18 | 343 | 89 | 73 | 56 | 28 | 10 |
| map_beam_none | F-M-Hm | test | 1,917 | 1,917 | 2 | 1,314 | 210 | 94 | 72 | 21 | 10 |
| map_beam_none | F-M-Hm | few-shot | 1,959 | 1,959 | 0 | 1,527 | 461 | 139 | 71 | 17 | 10 |
| elastic_pathing_hp | F-M-Hp | test | 1,917 | 1,828 | 1,402 | 87 | 77 | 76 | 75 | 84 | 1 |
| elastic_pathing_hp | F-M-Hp | few-shot | 1,959 | 1,881 | 1,488 | 80 | 66 | 65 | 64 | 85 | 1 |
| elastic_pathing | F-M-Hm | test | 1,917 | 1,855 | 1,484 | 129 | 103 | 98 | 96 | 87 | 1 |
| elastic_pathing | F-M-Hm | few-shot | 1,959 | 1,918 | 1,577 | 177 | 114 | 111 | 111 | 87 | 1 |
| can_trace | F-M-Hm | test | 245 | 2 | 0 | fail | fail | fail | fail | 50 | 2 |
| can_trace | F-M-Hm | few-shot | 162 | 5 | 2 | fail | fail | fail | fail | 80 | 2 |
| can_trace_area | F-M-Hm | test | 245 | 85 | 14 | fail | fail | fail | fail | 36 | 6 |
| can_trace_area | F-M-Hm | few-shot | 162 | 64 | 5 | fail | fail | fail | fail | 27 | 6 |
| dewri_hp | F-M-Hp | test | 1,917 | 1,899 | 347 | 81 | 58 | 51 | 45 | 50 | 9 |
| dewri_hp | F-M-Hp | few-shot | 1,959 | 1,905 | 311 | 89 | 49 | 41 | 35 | 47 | 6 |
| dewri | F-M-Hm | test | 1,917 | 1,909 | 40 | 293 | 70 | 60 | 49 | 39 | 10 |
| dewri | F-M-Hm | few-shot | 1,959 | 1,933 | 40 | 487 | 65 | 48 | 38 | 34 | 10 |

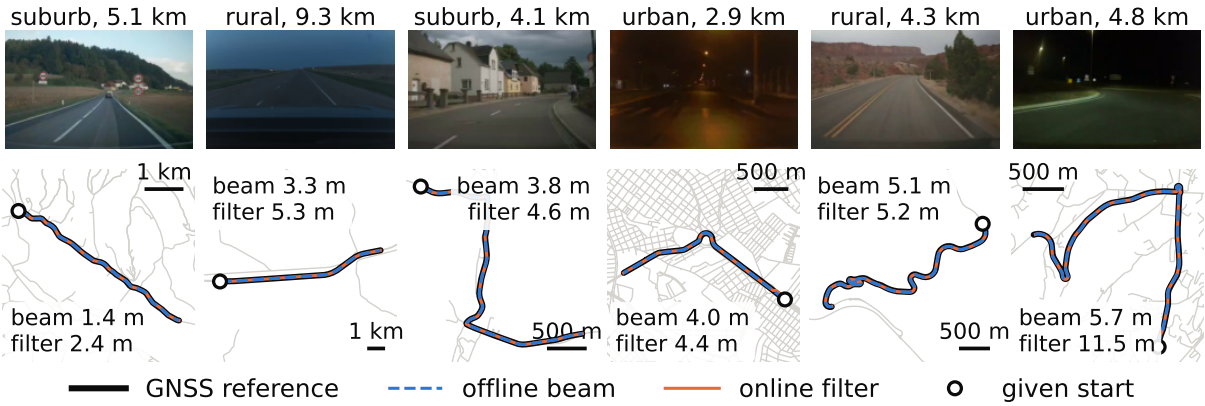



Fig. 3. Six held-out five-minute windows recovered by the offline beam and the online filter (speed, steering, start and heading known) below their camera frames at mid-window (not an input); grey: roads; numbers: ADE.

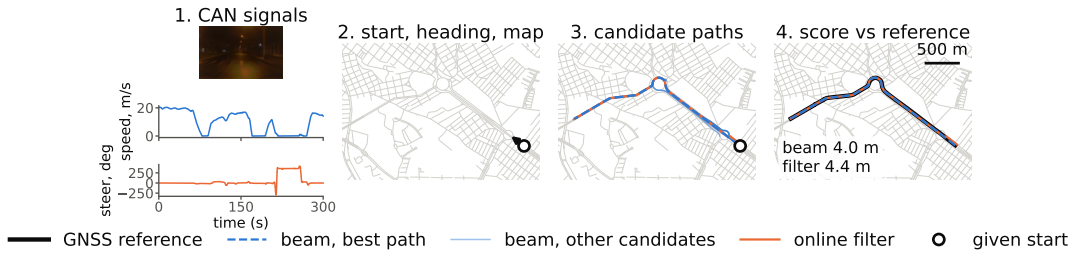



Fig. 4. The pipeline on one held-out window: CAN signals, start, heading and visible roads in; candidate paths; scoring at the reference seconds.

TABLE IX

FIXED MAP ATTACKS ON DEGRADED INPUTS (FIVE-MINUTE WINDOWS, GRID FIXED BEFORE THE FIRST RUN): MEDIAN ADE AND ROUTE RECOVERY ON THE CLEAN AND THE DEGRADED SIGNAL, AND THE PAIRED CHANGE OF THE ROUTE RECOVERY RATE IN PERCENTAGE POINTS WITH 95 % DRIVER-CLUSTERED CI.

| method | protocol | degradation | fold | $n$ | med. clean | med. degraded | route clean % | route degraded % | $\Delta$ route pp [CI] |
|---|---|---|---|---|---|---|---|---|---|
| map PF, steering | O-M-Hp | steering quantised to 2 deg | test | 1,917 | 43 | 43 | 87 | 87 | +0.0 [-0.2, +0.3] |
| map PF, steering | O-M-Hp | steering quantised to 2 deg | few-shot | 1,959 | 27 | 27 | 90 | 90 | -0.2 [-0.6, +0.0] |
| map PF, steering | O-M-Hp | steering quantised to 5 deg | test | 1,917 | 43 | 42 | 87 | 87 | +0.2 [-0.2, +0.7] |
| map PF, steering | O-M-Hp | steering quantised to 5 deg | few-shot | 1,959 | 27 | 26 | 90 | 90 | -0.3 [-0.8, +0.1] |
| map PF, steering | O-M-Hp | steering quantised to 10 deg | test | 1,917 | 43 | 42 | 87 | 87 | +0.4 [-0.1, +0.9] |
| map PF, steering | O-M-Hp | steering quantised to 10 deg | few-shot | 1,959 | 27 | 27 | 90 | 90 | -0.6 [-1.3, -0.0] |
| map PF, steering | O-M-Hp | steering quantised to 20 deg | test | 1,917 | 43 | 44 | 87 | 86 | -1.4 [-2.5, -0.4] |
| map PF, steering | O-M-Hp | steering quantised to 20 deg | few-shot | 1,959 | 27 | 28 | 90 | 89 | -1.1 [-2.3, -0.4] |
| map PF, steering | O-M-Hp | steering at 1 Hz | test | 1,917 | 43 | 42 | 87 | 88 | +0.9 [+0.4, +1.8] |
| map PF, steering | O-M-Hp | steering at 1 Hz | few-shot | 1,959 | 27 | 27 | 90 | 91 | +0.8 [+0.3, +1.3] |
| map PF, steering | O-M-Hp | steering at 0.2 Hz | test | 1,917 | 43 | 49 | 87 | 84 | -2.7 [-5.8, -1.4] |
| map PF, steering | O-M-Hp | steering at 0.2 Hz | few-shot | 1,959 | 27 | 38 | 90 | 85 | -5.6 [-7.8, -3.8] |
| map PF, steering | O-M-Hp | steering offset noise 2 deg | test | 1,917 | 43 | 42 | 87 | 87 | -0.1 [-0.8, +0.3] |
| map PF, steering | O-M-Hp | steering offset noise 2 deg | few-shot | 1,959 | 27 | 27 | 90 | 90 | -0.3 [-0.8, +0.2] |
| map PF, steering | O-M-Hp | steering offset noise 5 deg | test | 1,917 | 43 | 43 | 87 | 87 | -0.5 [-1.2, +0.0] |
| map PF, steering | O-M-Hp | steering offset noise 5 deg | few-shot | 1,959 | 27 | 27 | 90 | 89 | -0.8 [-1.7, -0.1] |
| map PF, steering | O-M-Hp | steering delayed 0.5 s | test | 1,917 | 43 | 42 | 87 | 88 | +1.1 [+0.4, +2.1] |
| map PF, steering | O-M-Hp | steering delayed 0.5 s | few-shot | 1,959 | 27 | 27 | 90 | 91 | +0.6 [-0.3, +1.2] |
| map PF, steering | O-M-Hp | speed quantised to 1 m/s | test | 1,917 | 43 | 42 | 87 | 87 | +0.1 [-0.5, +0.5] |
| map PF, steering | O-M-Hp | speed quantised to 1 m/s | few-shot | 1,959 | 27 | 31 | 90 | 89 | -0.7 [-1.4, +0.5] |
| map PF, steering | O-M-Hp | speed quantised to 2 m/s | test | 1,917 | 43 | 41 | 87 | 87 | -0.2 [-0.7, +0.6] |
| map PF, steering | O-M-Hp | speed quantised to 2 m/s | few-shot | 1,959 | 27 | 42 | 90 | 88 | -2.6 [-4.1, -0.7] |
| map PF, steering | O-M-Hp | speed quantised to 5 m/s | test | 1,917 | 43 | 122 | 87 | 83 | -4.4 [-7.3, -2.4] |
| map PF, steering | O-M-Hp | speed quantised to 5 m/s | few-shot | 1,959 | 27 | 68 | 90 | 83 | -7.6 [-9.7, -5.0] |
| map PF, steering | O-M-Hp | speed at 1 Hz | test | 1,917 | 43 | 44 | 87 | 86 | -0.9 [-1.9, -0.1] |
| map PF, steering | O-M-Hp | speed at 1 Hz | few-shot | 1,959 | 27 | 29 | 90 | 89 | -1.3 [-1.8, -0.5] |
| map beam, steering | F-M-Hp | steering quantised to 2 deg | test | 1,917 | 37 | 37 | 88 | 88 | +0.1 [+0.0, +0.2] |
| map beam, steering | F-M-Hp | steering quantised to 2 deg | few-shot | 1,959 | 21 | 21 | 92 | 92 | -0.1 [-0.2, +0.0] |
| map beam, steering | F-M-Hp | steering quantised to 5 deg | test | 1,917 | 37 | 36 | 88 | 88 | +0.2 [-0.1, +0.8] |
| map beam, steering | F-M-Hp | steering quantised to 5 deg | few-shot | 1,959 | 21 | 22 | 92 | 92 | -0.1 [-0.4, +0.2] |
| map beam, steering | F-M-Hp | steering quantised to 10 deg | test | 1,917 | 37 | 37 | 88 | 88 | +0.0 [-0.3, +0.4] |
| map beam, steering | F-M-Hp | steering quantised to 10 deg | few-shot | 1,959 | 21 | 22 | 92 | 92 | -0.3 [-0.7, +0.1] |
| map beam, steering | F-M-Hp | steering quantised to 20 deg | test | 1,917 | 37 | 39 | 88 | 86 | -1.4 [-2.4, -0.3] |
| map beam, steering | F-M-Hp | steering quantised to 20 deg | few-shot | 1,959 | 21 | 25 | 92 | 91 | -1.1 [-2.2, -0.4] |
| map beam, steering | F-M-Hp | steering at 1 Hz | test | 1,917 | 37 | 37 | 88 | 89 | +1.0 [+0.2, +2.0] |
| map beam, steering | F-M-Hp | steering at 1 Hz | few-shot | 1,959 | 21 | 22 | 92 | 93 | +1.3 [+0.8, +2.0] |
| map beam, steering | F-M-Hp | steering at 0.2 Hz | test | 1,917 | 37 | 46 | 88 | 89 | +1.0 [-0.8, +3.0] |
| map beam, steering | F-M-Hp | steering at 0.2 Hz | few-shot | 1,959 | 21 | 36 | 92 | 90 | -1.8 [-3.4, -0.1] |
| map beam, steering | F-M-Hp | steering offset noise 2 deg | test | 1,917 | 37 | 37 | 88 | 87 | -0.2 [-0.4, +0.0] |
| map beam, steering | F-M-Hp | steering offset noise 2 deg | few-shot | 1,959 | 21 | 22 | 92 | 92 | -0.2 [-0.4, +0.2] |
| map beam, steering | F-M-Hp | steering offset noise 5 deg | test | 1,917 | 37 | 37 | 88 | 87 | -0.6 [-1.4, -0.1] |
| map beam, steering | F-M-Hp | steering offset noise 5 deg | few-shot | 1,959 | 21 | 23 | 92 | 92 | -0.1 [-0.5, +0.5] |
| map beam, steering | F-M-Hp | steering delayed 0.5 s | test | 1,917 | 37 | 37 | 88 | 89 | +1.3 [+0.5, +2.5] |
| map beam, steering | F-M-Hp | steering delayed 0.5 s | few-shot | 1,959 | 21 | 22 | 92 | 94 | +1.6 [+0.8, +2.4] |
| map beam, steering | F-M-Hp | speed quantised to 1 m/s | test | 1,917 | 37 | 36 | 88 | 88 | +0.0 [-0.7, +0.6] |
| map beam, steering | F-M-Hp | speed quantised to 1 m/s | few-shot | 1,959 | 21 | 26 | 92 | 91 | -0.7 [-1.2, +0.3] |
| map beam, steering | F-M-Hp | speed quantised to 2 m/s | test | 1,917 | 37 | 37 | 88 | 87 | -0.1 [-0.7, +0.8] |
| map beam, steering | F-M-Hp | speed quantised to 2 m/s | few-shot | 1,959 | 21 | 34 | 92 | 90 | -2.0 [-3.6, +0.0] |
| map beam, steering | F-M-Hp | speed quantised to 5 m/s | test | 1,917 | 37 | 105 | 88 | 84 | -3.3 [-5.0, -2.0] |
| map beam, steering | F-M-Hp | speed quantised to 5 m/s | few-shot | 1,959 | 21 | 59 | 92 | 85 | -6.7 [-8.7, -3.9] |
| map beam, steering | F-M-Hp | speed at 1 Hz | test | 1,917 | 37 | 37 | 88 | 87 | -0.2 [-1.1, +0.5] |
| map beam, steering | F-M-Hp | speed at 1 Hz | few-shot | 1,959 | 21 | 23 | 92 | 91 | -1.2 [-1.8, -0.6] |
| map beam, speed only | F-M-Hp | speed quantised to 1 m/s | test | 1,917 | 165 | 169 | 54 | 54 | -0.1 [-0.3, +0.1] |
| map beam, speed only | F-M-Hp | speed quantised to 1 m/s | few-shot | 1,959 | 343 | 338 | 47 | 47 | +0.1 [-0.4, +0.7] |
| map beam, speed only | F-M-Hp | speed quantised to 2 m/s | test | 1,917 | 165 | 174 | 54 | 54 | -0.1 [-0.5, +0.3] |
| map beam, speed only | F-M-Hp | speed quantised to 2 m/s | few-shot | 1,959 | 343 | 336 | 47 | 47 | -0.2 [-0.6, +0.4] |
| map beam, speed only | F-M-Hp | speed quantised to 5 m/s | test | 1,917 | 165 | 247 | 54 | 53 | -0.6 [-1.5, +0.1] |
| map beam, speed only | F-M-Hp | speed quantised to 5 m/s | few-shot | 1,959 | 343 | 323 | 47 | 47 | +0.5 [-0.5, +1.1] |
| map beam, speed only | F-M-Hp | speed at 1 Hz | test | 1,917 | 165 | 169 | 54 | 54 | -0.1 [-0.3, +0.1] |
| map beam, speed only | F-M-Hp | speed at 1 Hz | few-shot | 1,959 | 343 | 343 | 47 | 47 | +0.1 [+0.0, +0.3] |

TABLE X

DEGRADED INPUTS FOR DEAD RECKONING (NO MAP): CLEAN AND DEGRADED MEDIANS AND THE PAIRED MEDIAN / MEAN DIFFERENCES (METRES).

| method | degradation | fold | med. clean | med. degraded | $n$ | paired median [CI] | paired mean [CI] |
|---|---|---|---|---|---|---|---|
| steering DR | speed quantised to 1 m/s | test | 921 | 920 | 1,917 | -0.1 [-0.3, 0.2] | 0 [-1, 1] |
| steering DR | speed quantised to 1 m/s | few-shot | 1,669 | 1,664 | 1,963 | -0.1 [-0.6, 0.1] | -0 [-1, 1] |
| steering DR | speed noise 0.5 m/s | test | 921 | 926 | 1,917 | 0.1 [-0.0, 0.2] | 1 [0, 2] |
| steering DR | speed noise 0.5 m/s | few-shot | 1,669 | 1,670 | 1,963 | 0.1 [0.0, 0.3] | 2 [1, 3] |
| steering DR | steering at 1 Hz | test | 921 | 934 | 1,917 | 0.4 [-1.3, 1.6] | 7 [-1, 14] |
| steering DR | steering at 1 Hz | few-shot | 1,669 | 1,683 | 1,963 | 0.2 [-1.0, 2.3] | 5 [-1, 15] |
| straight line | speed quantised to 1 m/s | test | 735 | 736 | 1,917 | 0.0 [-0.1, 0.2] | 0 [0, 1] |
| straight line | speed quantised to 1 m/s | few-shot | 825 | 824 | 1,963 | 0.1 [-0.1, 0.5] | 1 [-0, 2] |
| straight line | speed noise 0.5 m/s | test | 735 | 736 | 1,917 | 0.1 [0.0, 0.3] | 1 [1, 2] |
| straight line | speed noise 0.5 m/s | few-shot | 825 | 825 | 1,963 | 0.1 [0.1, 0.2] | 1 [1, 1] |
| straight line | steering at 1 Hz | test | 735 | 735 | 1,917 | 0.0 [0.0, 0.0] | 0 [0, 0] |
| straight line | steering at 1 Hz | few-shot | 825 | 825 | 1,963 | 0.0 [0.0, 0.0] | 0 [0, 0] |

TABLE XI

COST STUDY ON ONE FIXED VALIDATION SUBSET, ONE WORKER PER METHOD: INFERENCE SECONDS PER WINDOW (MEDIAN AND 90TH PERCENTILE; FAILED SEARCHES COUNT WITH THE TIME THEY RAN), PEAK MEMORY AND THE PER-DRIVE GRAPH BUILD.

| method | $n_5$ | fail | med s | p90 s | $n_{15}$ | fail | med s | p90 s | RSS GB | graph s |
|---|---|---|---|---|---|---|---|---|---|---|
| dewri | 76 | 0 | 0.04 | 0.09 | 16 | 1 | 0.33 | 0.83 | 1.8 | 2.7 |
| dewri_hp | 76 | 1 | 0.03 | 0.06 | 16 | 2 | 0.23 | 0.60 | 1.8 | 2.7 |
| dr_none | 76 | 0 | 0.02 | 0.02 | 16 | 0 | 0.06 | 0.06 | 0.8 | – |
| dr_steer | 76 | 0 | 0.03 | 0.03 | 16 | 0 | 0.09 | 0.09 | 0.8 | – |
| elastic_pathing | 76 | 0 | 0.01 | 0.13 | 16 | 0 | 0.15 | 0.54 | 1.8 | 2.6 |
| elastic_pathing_hp | 76 | 0 | 0.01 | 0.12 | 16 | 0 | 0.16 | 0.52 | 1.8 | 2.6 |
| learned_gru | 76 | 0 | 0.22 | 0.23 | 16 | 0 | 0.66 | 0.67 | 0.8 | – |
| macrotracking | 76 | 0 | 0.08 | 0.21 | 16 | 0 | 0.43 | 0.69 | 1.8 | 2.6 |
| map_beam_none | 76 | 0 | 0.12 | 0.18 | 16 | 0 | 0.43 | 0.50 | 1.8 | 2.6 |
| map_beam_steer | 76 | 0 | 0.24 | 0.51 | 16 | 0 | 0.93 | 1.33 | 1.8 | 2.6 |
| map_beam_steer_b50 | 76 | 0 | 0.07 | 0.13 | 16 | 0 | 0.25 | 0.37 | 1.8 | 2.6 |
| map_pf_learned | 76 | 0 | 0.83 | 0.95 | 16 | 0 | 2.56 | 2.89 | 1.7 | 2.6 |
| map_pf_none | 76 | 0 | 0.89 | 1.38 | 16 | 0 | 4.03 | 4.97 | 1.7 | 2.6 |
| map_pf_steer | 76 | 0 | 0.57 | 0.66 | 16 | 0 | 1.77 | 2.05 | 1.7 | 2.6 |
| map_pf_steer_n500 | 76 | 0 | 0.22 | 0.30 | 16 | 0 | 0.70 | 0.87 | 1.7 | 2.6 |
| opd2 | 76 | 0 | 0.08 | 0.23 | 16 | 0 | 0.19 | 0.28 | 2.1 | 2.6 |
| spy | 76 | 0 | 0.13 | 0.68 | 16 | 0 | 0.24 | 1.23 | 1.8 | 2.7 |
| spy_area | 76 | 9 | 4.93 | 13.53 | 16 | 1 | 7.71 | 23.97 | 1.8 | 2.7 |
| spy_hp | 76 | 0 | 0.08 | 0.32 | 16 | 0 | 0.22 | 0.62 | 1.8 | 2.7 |

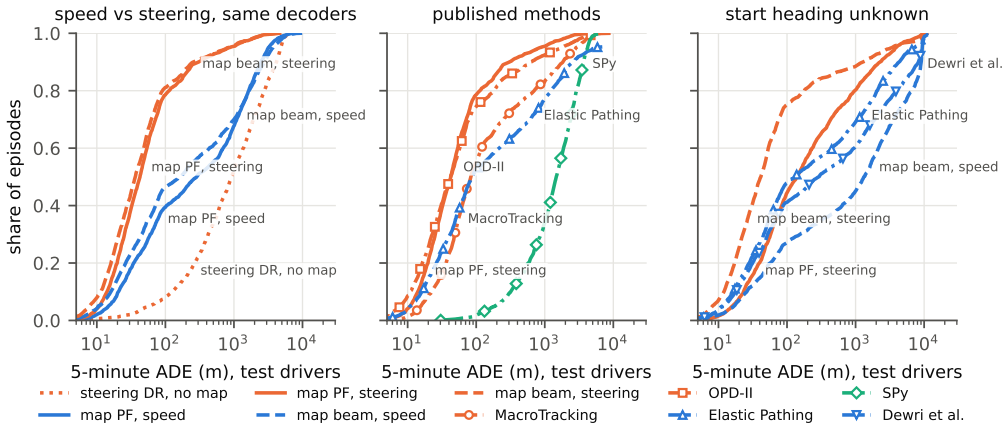



Fig. 5. ADE distributions of the main rows on five-minute windows.

TABLE XII

ACCURACY ON THE COST SUBSET AT THE PUBLISHED AND AT SMALLER SEARCH BUDGETS (VALIDATION WINDOWS).

| method / budget | protocol | window | $n$ | fail | median ADE m | route % |
|---|---|---|---|---|---|---|
| dewri__G1 | F-M-Hm | 5 min | 76 | 0 | 1,233 | 21 |
| dewri__G1 | F-M-Hm | 15 min | 16 | 1 | 4,170 | 12 |
| dewri_hp__G1 | F-M-Hp | 5 min | 76 | 1 | 524 | 34 |
| dewri_hp__G1 | F-M-Hp | 15 min | 16 | 2 | 1,737 | 25 |
| dr_none__G1 | O-N-Hp | 5 min | 76 | 0 | 497 | 16 |
| dr_none__G1 | O-N-Hp | 15 min | 16 | 0 | 3,132 | 6 |
| dr_steer__G3 | O-N-Hp | 5 min | 76 | 0 | 745 | 3 |
| dr_steer__G3 | O-N-Hp | 15 min | 16 | 0 | 4,681 | 0 |
| elastic_pathing__G1 | F-M-Hm | 5 min | 76 | 0 | 428 | 38 |
| elastic_pathing__G1 | F-M-Hm | 15 min | 16 | 0 | 3,268 | 19 |
| elastic_pathing_hp__G1 | F-M-Hp | 5 min | 76 | 0 | 205 | 46 |
| elastic_pathing_hp__G1 | F-M-Hp | 15 min | 16 | 0 | 2,100 | 25 |
| learned_gru_s0__G3 | O-N-Hp | 5 min | 76 | 0 | 203 | 12 |
| learned_gru_s0__G3 | O-N-Hp | 15 min | 16 | 0 | 1,237 | 0 |
| macrotracking__G3 | O-M-Hp | 5 min | 76 | 0 | 75 | 50 |
| macrotracking__G3 | O-M-Hp | 15 min | 16 | 0 | 1,709 | 25 |
| map_beam_none__G1 | F-M-Hp | 5 min | 76 | 0 | 549 | 32 |
| map_beam_none__G1 | F-M-Hp | 15 min | 16 | 0 | 4,260 | 12 |
| map_beam_steer__G3 | F-M-Hp | 5 min | 76 | 0 | 21 | 84 |
| map_beam_steer__G3 | F-M-Hp | 15 min | 16 | 0 | 58 | 81 |
| map_beam_steer_b50__G3 | F-M-Hp | 5 min | 76 | 0 | 21 | 84 |
| map_beam_steer_b50__G3 | F-M-Hp | 15 min | 16 | 0 | 58 | 81 |
| map_pf_learned_gru_s0__G3 | O-M-Hp | 5 min | 76 | 0 | 38 | 84 |
| map_pf_learned_gru_s0__G3 | O-M-Hp | 15 min | 16 | 0 | 115 | 88 |
| map_pf_none__G1 | O-M-Hp | 5 min | 76 | 0 | 933 | 36 |
| map_pf_none__G1 | O-M-Hp | 15 min | 16 | 0 | 3,935 | 19 |
| map_pf_steer__G3 | O-M-Hp | 5 min | 76 | 0 | 26 | 87 |
| map_pf_steer__G3 | O-M-Hp | 15 min | 16 | 0 | 90 | 81 |
| map_pf_steer_n500__G3 | O-M-Hp | 5 min | 76 | 0 | 24 | 87 |
| map_pf_steer_n500__G3 | O-M-Hp | 15 min | 16 | 0 | 91 | 81 |
| opd2__G3 | F-M-Hp | 5 min | 76 | 0 | 23 | 76 |
| opd2__G3 | F-M-Hp | 15 min | 16 | 0 | 1,317 | 31 |
| spy__G3 | F-M-Hm | 5 min | 76 | 0 | 1,306 | 7 |
| spy__G3 | F-M-Hm | 15 min | 16 | 0 | 5,401 | 6 |
| spy_area__G3 | F-M-Hm | 5 min | 76 | 9 | 2,743 | 0 |
| spy_area__G3 | F-M-Hm | 15 min | 16 | 1 | 6,180 | 0 |
| spy_hp__G3 | F-M-Hp | 5 min | 76 | 0 | 965 | 16 |
| spy_hp__G3 | F-M-Hp | 15 min | 16 | 0 | 4,946 | 6 |

TABLE XIII

CANDIDATE CONFIDENCE SCORE OF THE OFFLINE BEAM: AUROC OF THE TOP-1 SCORE FOR ROUTE RECOVERY (DRIVER-CLUSTERED CI), SPEARMAN CORRELATION WITH THE ADE, AND THE SHARE OF WINDOWS WHOSE SCORE IS 1.

| method | protocol | window | fold | $n$ | fail | AUROC [CI] | Spearman | score = 1 % |
|---|---|---|---|---|---|---|---|---|
| map_beam_steer | F-M-Hp | 5 min | test | 1,917 | 0 | 0.47 [0.41, 0.52] | -0.01 | 29 |
| map_beam_steer | F-M-Hp | 5 min | few-shot | 1,959 | 0 | 0.49 [0.44, 0.54] | -0.06 | 38 |
| map_beam_steer | F-M-Hp | 15 min | test | 608 | 0 | 0.54 [0.48, 0.59] | -0.08 | 19 |
| map_beam_steer | F-M-Hp | 15 min | few-shot | 561 | 0 | 0.54 [0.47, 0.59] | -0.20 | 34 |
| map_beam_steer | F-M-Hm | 5 min | test | 1,917 | 0 | 0.57 [0.53, 0.62] | -0.11 | 16 |
| map_beam_steer | F-M-Hm | 5 min | few-shot | 1,959 | 0 | 0.57 [0.51, 0.61] | -0.15 | 27 |
| map_beam_steer | F-M-Hm | 15 min | test | 608 | 0 | 0.55 [0.50, 0.60] | -0.09 | 15 |
| map_beam_steer | F-M-Hm | 15 min | few-shot | 561 | 0 | 0.58 [0.48, 0.65] | -0.26 | 30 |
| map_beam_none | F-M-Hp | 5 min | test | 1,917 | 0 | 0.81 [0.73, 0.86] | -0.51 | 7 |
| map_beam_none | F-M-Hp | 5 min | few-shot | 1,959 | 0 | 0.79 [0.74, 0.84] | -0.40 | 8 |
| map_beam_none | F-M-Hp | 15 min | test | 608 | 0 | 0.79 [0.63, 0.85] | -0.42 | 3 |
| map_beam_none | F-M-Hp | 15 min | few-shot | 561 | 0 | 0.77 [0.70, 0.82] | -0.26 | 2 |
| map_beam_none | F-M-Hm | 5 min | test | 1,917 | 0 | 0.65 [0.59, 0.67] | 0.02 | 1 |
| map_beam_none | F-M-Hm | 5 min | few-shot | 1,959 | 0 | 0.68 [0.65, 0.71] | 0.03 | 0 |
| map_beam_none | F-M-Hm | 15 min | test | 608 | 0 | 0.69 [0.56, 0.71] | 0.04 | 0 |
| map_beam_none | F-M-Hm | 15 min | few-shot | 561 | 0 | 0.69 [0.62, 0.73] | 0.07 | 0 |

TABLE XIV

SELECTIVE PREDICTION BY THE BEAM'S CONFIDENCE SCORE: WINDOWS ACCEPTED AT A THRESHOLD (COVERAGE, ALL EXPECTED WINDOWS IN THE DENOMINATOR), ROUTE RECOVERY AMONG THE ACCEPTED AND THE SHARE OF ACCEPTED ROUTES THAT ARE WRONG.

| method | protocol | window | fold | $\tau$ | coverage % | route among accepted % [CI] | wrong % |
|---|---|---|---|---|---|---|---|
| map_beam_steer | F-M-Hp | 5 min | test | 0.5 | 77 | 86 [80, 92] | 14 |
| map_beam_steer | F-M-Hp | 5 min | test | 0.7 | 57 | 86 [79, 92] | 14 |
| map_beam_steer | F-M-Hp | 5 min | test | 0.9 | 46 | 87 [81, 92] | 13 |
| map_beam_steer | F-M-Hp | 5 min | few-shot | 0.5 | 85 | 92 [88, 94] | 8 |
| map_beam_steer | F-M-Hp | 5 min | few-shot | 0.7 | 65 | 91 [86, 94] | 9 |
| map_beam_steer | F-M-Hp | 5 min | few-shot | 0.9 | 55 | 92 [87, 94] | 8 |
| map_beam_steer | F-M-Hp | 15 min | test | 0.5 | 84 | 73 [59, 83] | 27 |
| map_beam_steer | F-M-Hp | 15 min | test | 0.7 | 61 | 74 [59, 85] | 26 |
| map_beam_steer | F-M-Hp | 15 min | test | 0.9 | 45 | 76 [61, 85] | 24 |
| map_beam_steer | F-M-Hp | 15 min | few-shot | 0.5 | 89 | 76 [70, 81] | 24 |
| map_beam_steer | F-M-Hp | 15 min | few-shot | 0.7 | 69 | 76 [68, 80] | 24 |
| map_beam_steer | F-M-Hp | 15 min | few-shot | 0.9 | 54 | 76 [70, 81] | 24 |
| map_beam_steer | F-M-Hm | 5 min | test | 0.5 | 70 | 83 [77, 88] | 17 |
| map_beam_steer | F-M-Hm | 5 min | test | 0.7 | 50 | 84 [78, 89] | 16 |
| map_beam_steer | F-M-Hm | 5 min | test | 0.9 | 36 | 85 [79, 91] | 15 |
| map_beam_steer | F-M-Hm | 5 min | few-shot | 0.5 | 80 | 87 [84, 90] | 13 |
| map_beam_steer | F-M-Hm | 5 min | few-shot | 0.7 | 61 | 88 [84, 90] | 12 |
| map_beam_steer | F-M-Hm | 5 min | few-shot | 0.9 | 48 | 89 [85, 92] | 11 |
| map_beam_steer | F-M-Hm | 15 min | test | 0.5 | 82 | 72 [58, 83] | 28 |
| map_beam_steer | F-M-Hm | 15 min | test | 0.7 | 58 | 73 [58, 84] | 27 |
| map_beam_steer | F-M-Hm | 15 min | test | 0.9 | 42 | 74 [58, 86] | 26 |
| map_beam_steer | F-M-Hm | 15 min | few-shot | 0.5 | 88 | 75 [70, 81] | 25 |
| map_beam_steer | F-M-Hm | 15 min | few-shot | 0.7 | 69 | 75 [68, 79] | 25 |
| map_beam_steer | F-M-Hm | 15 min | few-shot | 0.9 | 53 | 79 [72, 83] | 21 |
| map_beam_none | F-M-Hp | 5 min | test | 0.5 | 28 | 89 [72, 95] | 11 |
| map_beam_none | F-M-Hp | 5 min | test | 0.7 | 19 | 94 [84, 97] | 6 |
| map_beam_none | F-M-Hp | 5 min | test | 0.9 | 13 | 97 [89, 99] | 3 |
| map_beam_none | F-M-Hp | 5 min | few-shot | 0.5 | 25 | 82 [73, 90] | 18 |
| map_beam_none | F-M-Hp | 5 min | few-shot | 0.7 | 15 | 91 [83, 96] | 9 |
| map_beam_none | F-M-Hp | 5 min | few-shot | 0.9 | 11 | 94 [88, 98] | 6 |
| map_beam_none | F-M-Hp | 15 min | test | 0.5 | 20 | 86 [51, 95] | 14 |
| map_beam_none | F-M-Hp | 15 min | test | 0.7 | 13 | 91 [64, 96] | 9 |
| map_beam_none | F-M-Hp | 15 min | test | 0.9 | 7 | 93 [60, 99] | 7 |
| map_beam_none | F-M-Hp | 15 min | few-shot | 0.5 | 16 | 67 [42, 81] | 33 |
| map_beam_none | F-M-Hp | 15 min | few-shot | 0.7 | 8 | 84 [55, 94] | 16 |
| map_beam_none | F-M-Hp | 15 min | few-shot | 0.9 | 5 | 93 [76, 100] | 7 |
| map_beam_none | F-M-Hm | 5 min | test | 0.5 | 17 | 50 [39, 55] | 50 |
| map_beam_none | F-M-Hm | 5 min | test | 0.7 | 6 | 56 [37, 69] | 44 |
| map_beam_none | F-M-Hm | 5 min | test | 0.9 | 2 | 75 [40, 90] | 25 |
| map_beam_none | F-M-Hm | 5 min | few-shot | 0.5 | 14 | 46 [42, 53] | 54 |
| map_beam_none | F-M-Hm | 5 min | few-shot | 0.7 | 5 | 58 [51, 70] | 42 |
| map_beam_none | F-M-Hm | 5 min | few-shot | 0.9 | 2 | 62 [49, 78] | 38 |
| map_beam_none | F-M-Hm | 15 min | test | 0.5 | 18 | 50 [23, 58] | 50 |
| map_beam_none | F-M-Hm | 15 min | test | 0.7 | 9 | 58 [34, 71] | 42 |
| map_beam_none | F-M-Hm | 15 min | test | 0.9 | 2 | 57 [8, 94] | 43 |
| map_beam_none | F-M-Hm | 15 min | few-shot | 0.5 | 10 | 42 [26, 54] | 58 |
| map_beam_none | F-M-Hm | 15 min | few-shot | 0.7 | 5 | 50 [33, 70] | 50 |
| map_beam_none | F-M-Hm | 15 min | few-shot | 0.9 | 2 | 36 [12, 57] | 64 |

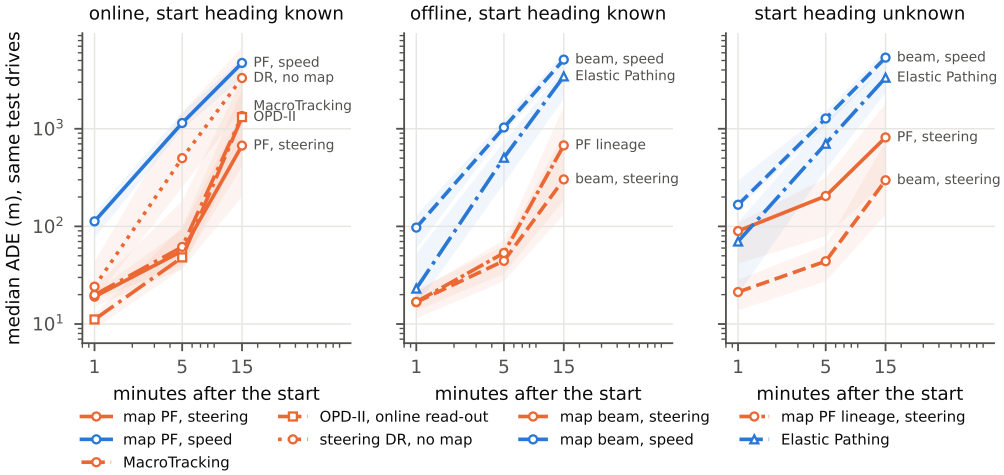



Fig. 6. Same-cohort horizon study: median ADE 1, 5 and 15 minutes after the start with driver-resampling bands.

TABLE XV

WINDOW TAXONOMY (FIVE MINUTES, WHOLE-PATH CRITERION AT THE GIVEN THRESHOLD): NO OUTPUT, NO CANDIDATE WITHIN THE THRESHOLD, A LOWER-RANKED CANDIDATE WITHIN IT WHILE THE TOP-1 IS NOT (RANKING FAILURE), TOP-1 WITHIN IT.

| method | protocol | fold | thr. m | $n$ | no output % | no cand. % | ranking % | Top-1 ok % |
|---|---|---|---|---|---|---|---|---|
| map_beam_steer | F-M-Hp | test | 50 | 1,917 | 0 | 28 | 10 | 61 |
| map_beam_steer | F-M-Hp | test | 100 | 1,917 | 0 | 12 | 7 | 81 |
| map_beam_steer | F-M-Hp | few-shot | 50 | 1,959 | 0 | 19 | 7 | 74 |
| map_beam_steer | F-M-Hp | few-shot | 100 | 1,959 | 0 | 6 | 6 | 88 |
| map_beam_steer | F-M-Hm | test | 50 | 1,917 | 0 | 29 | 15 | 57 |
| map_beam_steer | F-M-Hm | test | 100 | 1,917 | 0 | 13 | 12 | 75 |
| map_beam_steer | F-M-Hm | few-shot | 50 | 1,959 | 0 | 19 | 11 | 70 |
| map_beam_steer | F-M-Hm | few-shot | 100 | 1,959 | 0 | 7 | 10 | 83 |
| map_beam_none | F-M-Hp | test | 50 | 1,917 | 0 | 55 | 13 | 32 |
| map_beam_none | F-M-Hp | test | 100 | 1,917 | 0 | 41 | 13 | 46 |
| map_beam_none | F-M-Hp | few-shot | 50 | 1,959 | 0 | 53 | 18 | 29 |
| map_beam_none | F-M-Hp | few-shot | 100 | 1,959 | 0 | 40 | 18 | 42 |
| map_beam_none | F-M-Hm | test | 50 | 1,917 | 0 | 58 | 24 | 18 |
| map_beam_none | F-M-Hm | test | 100 | 1,917 | 0 | 45 | 28 | 27 |
| map_beam_none | F-M-Hm | few-shot | 50 | 1,959 | 0 | 57 | 27 | 16 |
| map_beam_none | F-M-Hm | few-shot | 100 | 1,959 | 0 | 45 | 30 | 25 |
| dewri_hp | F-M-Hp | test | 50 | 1,917 | 1 | 45 | 18 | 37 |
| dewri_hp | F-M-Hp | test | 100 | 1,917 | 1 | 25 | 20 | 54 |
| dewri_hp | F-M-Hp | few-shot | 50 | 1,959 | 3 | 37 | 22 | 39 |
| dewri_hp | F-M-Hp | few-shot | 100 | 1,959 | 3 | 22 | 23 | 52 |
| dewri | F-M-Hm | test | 50 | 1,917 | 0 | 49 | 22 | 29 |
| dewri | F-M-Hm | test | 100 | 1,917 | 0 | 30 | 28 | 41 |
| dewri | F-M-Hm | few-shot | 50 | 1,959 | 1 | 41 | 29 | 29 |
| dewri | F-M-Hm | few-shot | 100 | 1,959 | 1 | 27 | 32 | 39 |
| elastic_pathing_hp | F-M-Hp | test | 50 | 1,917 | 5 | 56 | 4 | 35 |
| elastic_pathing_hp | F-M-Hp | test | 100 | 1,917 | 5 | 40 | 3 | 52 |
| elastic_pathing_hp | F-M-Hp | few-shot | 50 | 1,959 | 4 | 52 | 5 | 39 |
| elastic_pathing_hp | F-M-Hp | few-shot | 100 | 1,959 | 4 | 37 | 5 | 54 |
| elastic_pathing | F-M-Hm | test | 50 | 1,917 | 3 | 61 | 3 | 33 |
| elastic_pathing | F-M-Hm | test | 100 | 1,917 | 3 | 46 | 2 | 48 |
| elastic_pathing | F-M-Hm | few-shot | 50 | 1,959 | 2 | 62 | 4 | 32 |
| elastic_pathing | F-M-Hm | few-shot | 100 | 1,959 | 2 | 49 | 4 | 45 |

TABLE XVI

COMPLEMENTARITY OF THE MAP METHODS (FIVE-MINUTE WINDOWS): PER-WINDOW ORACLES OVER A METHOD SET (LOWEST ADE; ANY ROUTE RECOVERED), THE SHARE OF WINDOWS EVERY MEMBER FAILS, AND THE FIXED REFERENCE METHOD.

| set | fold | $n$ | oracle median m [CI] | oracle route % [CI] | all fail % [CI] | reference | ref. median | ref. route % |
|---|---|---|---|---|---|---|---|---|
| A steering map, heading known | test | 1,917 | 25 [18, 35] | 97 [95, 98] | 3.1 [2.1, 5.2] | beam | 37 | 88 |
| A steering map, heading known | few-shot | 1,959 | 15 [11, 28] | 98 [97, 99] | 1.9 [1.2, 2.8] | beam | 21 | 92 |
| B speed-only map, heading known | test | 1,917 | 50 [25, 93] | 78 [68, 86] | 22.3 [14.3, 32.5] | beam_none | 165 | 54 |
| B speed-only map, heading known | few-shot | 1,959 | 40 [21, 67] | 77 [70, 82] | 22.8 [18.2, 29.9] | beam_none | 343 | 47 |
| B' speed-only map, heading unknown | test | 1,917 | 61 [29, 139] | 72 [59, 81] | 28.4 [19.0, 40.5] | beam_none_hm | 1,314 | 32 |
| B' speed-only map, heading unknown | few-shot | 1,959 | 54 [36, 81] | 70 [64, 74] | 30.1 [25.7, 36.2] | beam_none_hm | 1,527 | 29 |
| C published, heading known | test | 1,917 | 30 [21, 46] | 94 [90, 95] | 6.3 [4.9, 9.5] | nan | – | – |
| C published, heading known | few-shot | 1,959 | 17 [13, 36] | 95 [93, 97] | 4.7 [3.4, 7.5] | nan | – | – |
| D all map, heading known | test | 1,917 | 24 [18, 34] | 98 [96, 99] | 2.3 [1.4, 3.6] | beam | 37 | 88 |
| D all map, heading known | few-shot | 1,959 | 14 [11, 26] | 98 [97, 99] | 1.6 [0.9, 2.5] | beam | 21 | 92 |
| P pedal subset, heading unknown | test | 245 | 40 [29, 70] | 95 [86, 97] | 4.9 [2.7, 14.1] | beam_hm | 46 | 88 |
| P pedal subset, heading unknown | few-shot | 162 | 19 [14, 35] | 93 [87, 98] | 7.4 [2.4, 13.3] | beam_hm | 21 | 86 |

TABLE XVII

COMMON FAILURE OF THE STEERING AND SPEED MAP METHODS BY REFERENCE TURN COUNT AND ROAD DENSITY (FOLDS POOLED).

| stratum | level | $n$ | drivers | all fail % [CI] | oracle route % | beam route % |
|---|---|---|---|---|---|---|
| reference turns | 0-1 | 2,234 | 81 | 0.1 [0.0, 0.2] | 100 | 98 |
| reference turns | 2-3 | 712 | 72 | 1.4 [0.6, 2.9] | 99 | 88 |
| reference turns | 4-6 | 528 | 74 | 5.3 [3.3, 8.7] | 95 | 73 |
| reference turns | 7+ | 402 | 66 | 8.7 [5.3, 12.4] | 91 | 67 |
| density | low (56/246 per km2 cuts) | 1,292 | 33 | 0.9 [0.2, 1.7] | 99 | 97 |
| density | mid (56/246 per km2 cuts) | 1,292 | 82 | 1.9 [1.2, 2.8] | 98 | 87 |
| density | high (56/246 per km2 cuts) | 1,292 | 81 | 2.9 [1.6, 4.5] | 97 | 86 |
| all | all | 3,876 | 92 | 1.9 [1.4, 2.6] | 98 | 90 |

TABLE XVIII

PER-DRIVER ROUTE RECOVERY SHARE (FIVE-MINUTE WINDOWS; DRIVERS WITH AT LEAST FIVE WINDOWS): POOLED RATE, QUANTILES OVER DRIVERS AND THE SHARE OF DRIVERS BELOW 50 %.

| method | fold | windows | drivers | $\geq 5$ | pooled % | q10 | q25 | q50 | q75 | q90 | below 50 % |
|---|---|---|---|---|---|---|---|---|---|---|---|
| pf | test | 1,917 | 42 | 36 | 87 | 62 | 77 | 87 | 96 | 100 | 3 |
| pf | few-shot | 1,959 | 50 | 41 | 90 | 73 | 78 | 92 | 96 | 100 | 0 |
| beam | test | 1,917 | 42 | 36 | 88 | 62 | 75 | 88 | 100 | 100 | 3 |
| beam | few-shot | 1,959 | 50 | 41 | 92 | 77 | 83 | 94 | 97 | 100 | 0 |
| opd2 | test | 1,917 | 42 | 36 | 79 | 54 | 64 | 78 | 84 | 89 | 8 |
| opd2 | few-shot | 1,959 | 50 | 41 | 78 | 54 | 67 | 79 | 88 | 91 | 5 |
| macro | test | 1,917 | 42 | 36 | 62 | 21 | 33 | 50 | 73 | 82 | 50 |
| macro | few-shot | 1,959 | 50 | 41 | 65 | 32 | 42 | 61 | 77 | 84 | 37 |
| ep_hp | test | 1,917 | 42 | 36 | 58 | 23 | 31 | 41 | 64 | 71 | 61 |
| ep_hp | few-shot | 1,959 | 50 | 41 | 59 | 30 | 36 | 50 | 67 | 71 | 46 |
| dewri_hp | test | 1,917 | 42 | 36 | 62 | 15 | 26 | 45 | 61 | 78 | 53 |
| dewri_hp | few-shot | 1,959 | 50 | 41 | 59 | 17 | 30 | 56 | 70 | 85 | 49 |
| spy_hp | test | 1,917 | 42 | 36 | 20 | 0 | 0 | 10 | 18 | 33 | 100 |
| spy_hp | few-shot | 1,959 | 50 | 41 | 22 | 0 | 4 | 14 | 27 | 35 | 100 |
| pf_none | test | 1,917 | 42 | 36 | 56 | 9 | 22 | 35 | 58 | 66 | 61 |
| pf_none | few-shot | 1,959 | 50 | 41 | 48 | 12 | 29 | 50 | 64 | 75 | 49 |
| beam_none | test | 1,917 | 42 | 36 | 54 | 8 | 20 | 31 | 57 | 67 | 64 |
| beam_none | few-shot | 1,959 | 50 | 41 | 47 | 15 | 30 | 50 | 62 | 71 | 49 |
| ep | test | 1,917 | 42 | 36 | 53 | 14 | 25 | 34 | 51 | 69 | 72 |
| ep | few-shot | 1,959 | 50 | 41 | 50 | 23 | 32 | 43 | 57 | 67 | 59 |
| dewri | test | 1,917 | 42 | 36 | 46 | 9 | 17 | 35 | 50 | 63 | 72 |
| dewri | few-shot | 1,959 | 50 | 41 | 43 | 17 | 21 | 38 | 47 | 63 | 78 |

TABLE XIX

ROAD-DENSITY STRATA AROUND THE START (TERTILES; FOLDS POOLED): MEDIAN ADE WITH CI, 100 M COVERAGE AND TAIL-LOST SHARE.

| method | protocol | window | tertile | $n$ | drivers | median ADE m [CI] | c100 % | tail lost % |
|---|---|---|---|---|---|---|---|---|
| map_pf_steer__G3 | O-M-Hp | 5 min | low (¡ 56/km2) | 1,292 | 33 | 29 [18, 59] | 90 | 17 |
| map_pf_steer__G3 | O-M-Hp | 5 min | mid (56 to 246/km2) | 1,292 | 82 | 39 [27, 48] | 83 | 27 |
| map_pf_steer__G3 | O-M-Hp | 5 min | high (¿ 246/km2) | 1,292 | 81 | 36 [26, 45] | 83 | 27 |
| map_pf_steer__G3 | O-M-Hp | 15 min | low (¡ 56/km2) | 414 | 27 | 89 [46, 185] | 65 | 53 |
| map_pf_steer__G3 | O-M-Hp | 15 min | mid (56 to 246/km2) | 381 | 65 | 202 [132, 373] | 50 | 70 |
| map_pf_steer__G3 | O-M-Hp | 15 min | high (¿ 246/km2) | 374 | 61 | 264 [169, 564] | 51 | 67 |
| map_beam_steer__G3 | F-M-Hp | 5 min | low (¡ 56/km2) | 1,292 | 33 | 24 [16, 53] | 91 | 17 |
| map_beam_steer__G3 | F-M-Hp | 5 min | mid (56 to 246/km2) | 1,292 | 82 | 32 [23, 42] | 85 | 25 |
| map_beam_steer__G3 | F-M-Hp | 5 min | high (¿ 246/km2) | 1,292 | 81 | 33 [22, 40] | 84 | 27 |
| map_beam_steer__G3 | F-M-Hp | 15 min | low (¡ 56/km2) | 414 | 27 | 77 [44, 167] | 67 | 53 |
| map_beam_steer__G3 | F-M-Hp | 15 min | mid (56 to 246/km2) | 381 | 65 | 163 [105, 326] | 53 | 70 |
| map_beam_steer__G3 | F-M-Hp | 15 min | high (¿ 246/km2) | 374 | 61 | 162 [100, 275] | 57 | 61 |
| map_pf_none__G1 | O-M-Hp | 5 min | low (¡ 56/km2) | 1,292 | 33 | 73 [32, 537] | 73 | 37 |
| map_pf_none__G1 | O-M-Hp | 5 min | mid (56 to 246/km2) | 1,292 | 82 | 677 [234, 953] | 49 | 62 |
| map_pf_none__G1 | O-M-Hp | 5 min | high (¿ 246/km2) | 1,292 | 81 | 843 [627, 978] | 40 | 70 |
| map_pf_none__G1 | O-M-Hp | 15 min | low (¡ 56/km2) | 414 | 27 | 1,023 [96, 5,289] | 42 | 77 |
| map_pf_none__G1 | O-M-Hp | 15 min | mid (56 to 246/km2) | 381 | 65 | 3,457 [2,459, 4,268] | 23 | 86 |
| map_pf_none__G1 | O-M-Hp | 15 min | high (¿ 246/km2) | 374 | 61 | 3,468 [2,920, 4,276] | 19 | 86 |
| dr_steer__G3 | O-N-Hp | 5 min | low (¡ 56/km2) | 1,296 | 33 | 2,102 [1,067, 4,256] | 15 | 99 |
| dr_steer__G3 | O-N-Hp | 5 min | mid (56 to 246/km2) | 1,292 | 82 | 1,245 [819, 1,859] | 22 | 95 |
| dr_steer__G3 | O-N-Hp | 5 min | high (¿ 246/km2) | 1,292 | 81 | 632 [428, 1,132] | 30 | 90 |
| dr_steer__G3 | O-N-Hp | 15 min | low (¡ 56/km2) | 418 | 27 | 8,506 [6,751, 12,469] | 5 | 99 |
| dr_steer__G3 | O-N-Hp | 15 min | mid (56 to 246/km2) | 381 | 65 | 6,461 [4,875, 7,873] | 9 | 99 |
| dr_steer__G3 | O-N-Hp | 15 min | high (¿ 246/km2) | 374 | 61 | 3,373 [1,772, 5,018] | 13 | 97 |
| dr_curvature__G5 | O-N-Hp | 5 min | low (¡ 56/km2) | 1,296 | 33 | 85 [74, 109] | 65 | 67 |
| dr_curvature__G5 | O-N-Hp | 5 min | mid (56 to 246/km2) | 1,292 | 82 | 86 [76, 102] | 63 | 63 |
| dr_curvature__G5 | O-N-Hp | 5 min | high (¿ 246/km2) | 1,292 | 81 | 70 [64, 82] | 69 | 55 |
| dr_curvature__G5 | O-N-Hp | 15 min | low (¡ 56/km2) | 418 | 27 | 516 [479, 724] | 25 | 94 |
| dr_curvature__G5 | O-N-Hp | 15 min | mid (56 to 246/km2) | 381 | 65 | 548 [443, 675] | 25 | 95 |
| dr_curvature__G5 | O-N-Hp | 15 min | high (¿ 246/km2) | 374 | 61 | 418 [350, 492] | 31 | 89 |

TABLE XX

EXTERNAL CHECK ON THE CONTROLLED SUPPLEMENT (ONE VEHICLE, ONE METROPOLITAN AREA, OTHER DRIVERS; FIVE-MINUTE WINDOWS), METHODS UNCHANGED AFTER FREEZING.

| method | protocol | $n$ | drives | drivers | fail | median ADE m [CI] | p90 m | route % [CI] |
|---|---|---|---|---|---|---|---|---|
| steering DR, no map | O-N-Hp | 46 | 27 | 12 | 0 | 117 [93, 170] | 449 | 7 [0, 10] |
| map PF, steering | O-M-Hp | 46 | 27 | 12 | 0 | 75 [46, 492] | 896 | 59 [26, 69] |
| MacroTracking | O-M-Hp | 46 | 27 | 12 | 0 | 19 [14, 24] | 144 | 85 [74, 95] |
| map beam, steering | F-M-Hp | 46 | 27 | 12 | 0 | 46 [34, 308] | 846 | 61 [37, 67] |
| OPD-II | F-M-Hp | 46 | 27 | 12 | 0 | 25 [11, 270] | 600 | 65 [32, 75] |
| map PF, speed | O-M-Hp | 46 | 27 | 12 | 0 | 762 [503, 1,011] | 1,545 | 9 [0, 20] |
| map beam, speed | F-M-Hp | 46 | 27 | 12 | 0 | 763 [321, 890] | 1,653 | 4 [0, 14] |
| Elastic Pathing, heading | F-M-Hp | 46 | 27 | 12 | 0 | 337 [239, 779] | 1,185 | 28 [6, 38] |
| Dewri et al., heading | F-M-Hp | 46 | 27 | 12 | 0 | 605 [345, 670] | 1,621 | 15 [0, 27] |

TABLE XXI

MAP-EXTENT CHECKS: SHARE OF WINDOWS WHOSE REFERENCE ROUTE LEAVES THE 5 KM AREA ADAPTATION SQUARE OR THE VISIBLE DISC.

| check | method | window | fold | $n$ | out | share % | median max m |
|---|---|---|---|---|---|---|---|
| out_square | ct | 5 min | test | 245 | 181 | 73.9 | 4,486 |
| out_square | ct | 5 min | few-shot | 162 | 106 | 65.4 | 3,822 |
| out_square | spy | 5 min | test | 1,917 | 1,531 | 79.9 | 5,728 |
| out_square | spy | 5 min | few-shot | 1,959 | 1,544 | 78.8 | 4,702 |
| out_disc | G3 | 5 min | test | 1,917 | 0 | 0.0 | 6,311 |
| out_disc | G3 | 5 min | few-shot | 1,959 | 0 | 0.0 | 5,168 |
| out_disc | G3 | 15 min | test | 608 | 0 | 0.0 | 17,222 |
| out_disc | G3 | 15 min | few-shot | 561 | 0 | 0.0 | 14,538 |
| out_disc | G3 | whole drive | test | 282 | 18 | 6.4 | 10,190 |
| out_disc | G3 | whole drive | few-shot | 454 | 37 | 8.1 | 6,263 |
| out_disc | G1 | 5 min | test | 1,917 | 0 | 0.0 | 6,311 |
| out_disc | G1 | 5 min | few-shot | 1,959 | 0 | 0.0 | 5,168 |
| out_disc | G1 | 15 min | test | 608 | 0 | 0.0 | 17,222 |
| out_disc | G1 | 15 min | few-shot | 561 | 0 | 0.0 | 14,538 |
| out_disc | G1 | whole drive | test | 282 | 18 | 6.4 | 10,190 |
| out_disc | G1 | whole drive | few-shot | 454 | 37 | 8.1 | 6,263 |

TABLE XXII

AGENT STUDY (SUPPLEMENTARY COHORT, ONE FIVE-MINUTE WINDOW PER HELD-OUT DRIVE): EVERY CONFIGURATION AND SEED, WITH THE SELECTOR FALLBACK ROW AND THE POOL ORACLE.

| row | fold | $n$ | route % [CI] | median ADE m [CI] | no out. % | cost s | calls | tokens |
|---|---|---|---|---|---|---|---|---|
| map beam (A) | test | 268 | 69 [58, 85] | 49 [29, 66] | 0.0 | 0.3 | – | – |
| supervised selector (B) | test | 268 | 75 [67, 87] | 38 [29, 50] | 0.0 | 1.1 | 3.0 | – |
| Qwen agent, seed 0 | test | 268 | 75 [66, 88] | 41 [29, 55] | 0.0 | 35.5 | 2.9 | 1,703 |
| Qwen agent, seed 1 | test | 268 | 75 [65, 87] | 44 [27, 54] | 0.0 | 37.8 | 2.9 | 1,753 |
| Qwen agent, seed 2 | test | 268 | 75 [66, 88] | 39 [29, 52] | 0.0 | 35.9 | 3.0 | 1,702 |
| Qwen agent (C), seeds pooled | test | 268 | 75 [66, 88] | 41 [29, 53] | 0.0 | 36.4 | 2.9 | 1,719 |
| Qwen agent with selector fallback | test | 268 | 75 [66, 88] | 41 [29, 53] | 0.0 | – | – | – |
| pool oracle (diagnostic) | test | 268 | 83 [74, 93] | 27 [19, 32] | 0.0 | – | – | – |
| map beam (A) | few-shot | 391 | 83 [76, 90] | 24 [18, 36] | 0.0 | 0.3 | – | – |
| supervised selector (B) | few-shot | 391 | 83 [78, 90] | 24 [18, 34] | 0.0 | 1.0 | 3.0 | – |
| Qwen agent, seed 0 | few-shot | 391 | 84 [77, 91] | 23 [18, 34] | 0.0 | 31.9 | 3.0 | 1,626 |
| Qwen agent, seed 1 | few-shot | 391 | 84 [77, 90] | 23 [18, 34] | 0.0 | 37.9 | 2.9 | 1,741 |
| Qwen agent, seed 2 | few-shot | 391 | 84 [78, 91] | 22 [18, 35] | 0.0 | 35.2 | 2.9 | 1,711 |
| Qwen agent (C), seeds pooled | few-shot | 391 | 84 [77, 91] | 23 [18, 34] | 0.0 | 35.0 | 2.9 | 1,693 |
| Qwen agent with selector fallback | few-shot | 391 | 84 [77, 91] | 23 [18, 34] | 0.0 | – | – | – |
| pool oracle (diagnostic) | few-shot | 391 | 92 [87, 96] | 16 [11, 23] | 0.0 | – | – | – |

TABLE XXIII

AGENT STUDY, PAIRED COMPARISONS ON THE SAME WINDOWS: ROUTE RECOVERY DIFFERENCE IN PERCENTAGE POINTS (SEEDS POOLED INSIDE THE DRIVER RESAMPLES; PER-SEED VALUES) AND THE MEDIAN ADE DIFFERENCE ON WINDOWS WITH OUTPUT ON BOTH SIDES.

| pair | fold | $n$ | Δ route pp [CI] | per seed | Δ median ADE m [CI] |
|---|---|---|---|---|---|
| agent_vs_selector | test | 268 | +0.1 [-1.6, +3.4] | +0.4, -0.4, +0.4 | 0.0 [0.0, 0.0] |
| agent_vs_beam | test | 268 | +5.7 [+1.6, +8.5] | +6.0, +5.2, +6.0 | 0.0 [0.0, 0.0] |
| agent_vs_selector | few-shot | 391 | +0.7 [-2.6, +2.8] | +0.8, +0.5, +0.8 | 0.0 [0.0, 0.0] |
| agent_vs_beam | few-shot | 391 | +1.2 [-1.3, +3.8] | +1.3, +1.0, +1.3 | 0.0 [0.0, 0.0] |

TABLE XXIV
SUPERVISED SELECTOR ON THE VALIDATION COHORT (CROSS-FITTED BY DRIVER).

| row | windows | drivers | route % [CI] |
|---|---|---|---|
| selector_cv | 610 | 43 | 82 [77, 87] |
| beam1 | 610 | 43 | 79 [72, 84] |
| pf1 | 610 | 43 | 75 [69, 80] |
| ep1 | 610 | 43 | 28 [21, 40] |
| oracle | 610 | 43 | 91 [88, 94] |

TABLE XXV
VALIDATION PILOT OF THE AGENT (40 WINDOWS, THREE SEEDS).

| tag | $n$ | valid % | trunc. % | forced % | turns | calls | tokens | thinking | LLM s |
|---|---|---|---|---|---|---|---|---|---|
| agent_qwen3_s0 | 40 | 100.0 | 2.5 | 87.5 | 3.8 | 3.0 | 1,703 | 1,521 | 27.9 |
| agent_qwen3_s1 | 40 | 100.0 | 0.0 | 85.0 | 3.8 | 2.9 | 1,754 | 1,577 | 38.0 |
| agent_qwen3_s2 | 40 | 100.0 | 0.0 | 77.5 | 3.7 | 2.9 | 1,693 | 1,513 | 35.6 |
| all | 120 | 100.0 | 0.8 | 83.3 | 3.8 | 2.9 | 1,717 | 1,537 | 34.6 |

TABLE XXVI
CEILING EXPLORATION OF THE AGENT (C-MAX: TEN RECONSTRUCTOR PRESETS, COMPARISON AND SCORING TOOLS, RELAXED BUDGETS, OPTIONAL CRITIC AND SEED VOTE); EXPLORATORY, NOT PART OF THE FROZEN PROTOCOL.

| row | fold | $n$ | route % [CI] | median ADE m [CI] | no out. % | cost s | calls | tokens |
|---|---|---|---|---|---|---|---|---|
| map beam (A) | test | 268 | 69 [58, 85] | 49 [29, 66] | 0.0 | 0.3 | – | – |
| supervised selector (B) | test | 268 | 74 [66, 85] | 43 [28, 59] | 0.0 | 4.8 | 3.0 | – |
| Qwen agent, seed 0 | test | 268 | 77 [70, 88] | 41 [28, 52] | 0.4 | 116.6 | 4.8 | 3,448 |
| Qwen agent, seed 1 | test | 268 | 79 [72, 89] | 38 [27, 49] | 0.4 | 116.8 | 4.9 | 3,497 |
| Qwen agent, seed 2 | test | 268 | 78 [70, 89] | 39 [27, 46] | 0.0 | 73.7 | 4.8 | 3,420 |
| Qwen agent (C), seeds pooled | test | 268 | 78 [71, 88] | 39 [27, 49] | 0.2 | 102.4 | 4.8 | 3,455 |
| Qwen agent with selector fallback | test | 268 | 78 [71, 88] | 39 [27, 49] | 0.0 | – | – | – |
| pool oracle (diagnostic) | test | 268 | 87 [83, 94] | 24 [18, 29] | 0.0 | – | – | – |
| agent_qwen3max_vote | test | 268 | 79 [72, 89] | 37 [27, 45] | 0.0 | – | – | – |
| agent_qwen3maxcritic_s0 | test | 268 | 78 [71, 88] | 40 [27, 50] | 0.0 | – | – | – |
| agent of run icde_agent (seeds pooled) | test | 268 | 75 [66, 88] | 41 [29, 53] | 0.0 | – | – | – |
| map beam (A) | few-shot | 391 | 83 [76, 90] | 24 [18, 36] | 0.0 | 0.3 | – | – |
| supervised selector (B) | few-shot | 391 | 82 [75, 90] | 26 [20, 41] | 0.0 | 4.4 | 3.0 | – |
| Qwen agent, seed 0 | few-shot | 391 | 81 [75, 87] | 27 [20, 39] | 0.0 | 115.2 | 4.7 | 3,559 |
| Qwen agent, seed 1 | few-shot | 391 | 83 [76, 89] | 26 [20, 37] | 0.0 | 109.9 | 4.8 | 3,498 |
| Qwen agent, seed 2 | few-shot | 391 | 81 [75, 88] | 28 [21, 37] | 0.3 | 72.5 | 4.7 | 3,422 |
| Qwen agent (C), seeds pooled | few-shot | 391 | 81 [75, 88] | 27 [21, 37] | 0.1 | 99.2 | 4.8 | 3,493 |
| Qwen agent with selector fallback | few-shot | 391 | 81 [75, 88] | 27 [21, 37] | 0.0 | – | – | – |
| pool oracle (diagnostic) | few-shot | 391 | 94 [90, 97] | 15 [11, 22] | 0.0 | – | – | – |
| agent_qwen3max_vote | few-shot | 391 | 83 [76, 89] | 25 [20, 36] | 0.0 | – | – | – |
| agent_qwen3maxcritic_s0 | few-shot | 391 | 82 [76, 88] | 26 [20, 34] | 0.3 | – | – | – |
| agent of run icde_agent (seeds pooled) | few-shot | 391 | 84 [77, 91] | 23 [18, 34] | 0.0 | – | – | – |
| agent, both folds pooled | both folds | 659 | 80 [75, 85] | 31 [25, 39] | 0.2 | – | – | – |
| agent_prev, both folds pooled | both folds | 659 | 80 [74, 87] | 30 [22, 41] | 0.0 | – | – | – |
| selector, both folds pooled | both folds | 659 | 79 [73, 85] | 31 [25, 44] | 0.0 | – | – | – |
| beam, both folds pooled | both folds | 659 | 77 [69, 86] | 32 [23, 47] | 0.0 | – | – | – |

TABLE XXVII
CEILING EXPLORATION, PAIRED ROUTE-RECOVERY DIFFERENCES (PERCENTAGE POINTS, DRIVER-CLUSTERED CIS).

| pair | fold | $n$ | $\Delta$ route pp [CI] |
|---|---|---|---|
| agent_qwen3max_vote_vs_selector | test | 268 | +4.9 [+1.9, +7.4] |
| agent_qwen3max_vote_vs_beam | test | 268 | +9.3 [+2.3, +14.6] |
| agent_qwen3maxcritic_s0_vs_selector | test | 268 | +4.1 [+1.1, +6.8] |
| agent_qwen3maxcritic_s0_vs_beam | test | 268 | +8.6 [+2.0, +13.6] |
| agent_vs_selector | test | 268 | +4.1 [+1.3, +7.6] |
| agent_vs_beam | test | 268 | +8.6 [+1.9, +13.4] |
| agent_vs_prev | test | 268 | +2.9 [-0.7, +5.3] |
| agent_qwen3max_vote_vs_selector | few-shot | 391 | +0.5 [-1.8, +3.4] |
| agent_qwen3max_vote_vs_beam | few-shot | 391 | -0.3 [-3.4, +4.5] |
| agent_qwen3maxcritic_s0_vs_selector | few-shot | 391 | +0.0 [-2.0, +2.1] |
| agent_qwen3maxcritic_s0_vs_beam | few-shot | 391 | -0.8 [-3.6, +3.0] |
| agent_vs_selector | few-shot | 391 | -0.7 [-2.8, +1.6] |
| agent_vs_beam | few-shot | 391 | -1.4 [-4.8, +3.1] |
| agent_vs_prev | few-shot | 391 | -2.6 [-4.8, +0.3] |
| agent_vs_selector | both folds | 659 | +1.3 [-0.7, +3.2] |
| agent_vs_beam | both folds | 659 | +2.6 [-1.8, +7.3] |
| agent_vs_prev | both folds | 659 | -0.4 [-2.8, +2.0] |

TABLE XXVIII
CEILING EXPLORATION: THE SELECTOR OVER THE TEN PRESETS AND THE EXTENDED DIAGNOSTICS ON THE CACHED VALIDATION WINDOWS (CROSS-FITTED BY DRIVER).

| row | windows | drivers | route % [CI] |
|---|---|---|---|
| selector_cv | 428 | 22 | 79 [74, 84] |
| beam1 | 428 | 22 | 77 [65, 83] |
| pf1 | 428 | 22 | 72 [65, 77] |
| ep1 | 428 | 22 | 24 [18, 41] |
| oracle | 428 | 22 | 92 [90, 94] |

TABLE XXIX
CALIBRATION HINTS GIVEN TO THE C-MAX CONTROLLER (VALIDATION CANDIDATES: SHARE THAT RECOVERED THE ROUTE, BY TOOL TOP-1 AND BY DIAGNOSTIC BIN).

| diagnostic | bin | candidates | route % |
|---|---|---|---|
| tool_top1 | beam | 428 | 77 |
| tool_top1 | beam_loose | 428 | 77 |
| tool_top1 | beam_wide | 428 | 77 |
| tool_top1 | ep | 419 | 25 |
| tool_top1 | ep_hm | 426 | 22 |
| tool_top1 | pf | 428 | 72 |
| tool_top1 | pf_s1 | 428 | 73 |
| tool_top1 | pf_s2 | 428 | 72 |
| tool_top1 | pf_s3 | 428 | 74 |
| tool_top1 | pf_s4 | 428 | 73 |
| consensus_tools | 0 | 7,419 | 6 |
| consensus_tools | 1 | 1,385 | 8 |
| consensus_tools | 2-3 | 1,284 | 29 |
| consensus_tools | 4-99 | 11,070 | 79 |
| hd_abs_rms_deg | 0-10 | 400 | 99 |
| hd_abs_rms_deg | 10-20 | 1,457 | 92 |
| hd_abs_rms_deg | 20-40 | 4,779 | 74 |
| hd_abs_rms_deg | ¿=40 | 14,817 | 31 |
| jn_disagree | 0 | 7,006 | 56 |
| jn_disagree | 1 | 5,276 | 46 |
| jn_disagree | 2-99 | 9,171 | 38 |
| dr_dist_first60_m | 0-20 | 4,130 | 51 |
| dr_dist_first60_m | 20-50 | 6,845 | 53 |
| dr_dist_first60_m | 50-150 | 6,631 | 52 |
| dr_dist_first60_m | ¿=150 | 3,847 | 18 |
| turns_unmatched | 0 | 4,732 | 86 |
| turns_unmatched | 1 | 5,076 | 53 |
| turns_unmatched | 2-99 | 11,645 | 27 |